Host Communication Documentation

SECS/GEM

- Chapter 1 -

TOWA Corporation

▼Contents

1. General Requirements 4

2. State Models 5

2-1 Communications and Control State Model 5

2-2 Equipment Processing State Model 9

3. Gem Compliance Statement 10

4. Gem Scenario 11

4-1. Establish Communications 11

4-1-1 Host Attempts To Establish Communications 11
4-1-2 Equipment Attempts To Establish Communications and Host Acknowledges 12
4-1-3 Equipment Receives S1F14 From Host Before Sending S1F14 13
4-1-4 Equipment Sends S1F14 To Host Before Receiving S1F14 13

4-2. Event Notification 14

4-2-1 Collection Event which occurs on the Equipment 14
4-2-2 Host Requests Event Report 14

4-3. Dynamic Event Report Configuration 15

4-3-1 Collection Event Reporting Set-up 15

4-4. Variable Data Collection 16

4-4-1 Host Requests Report 16

4-5. Trace Data Collection 17

4-5-1 Host Initiates Trace Report : 17

4-6. Limits Monitoring 18

4-6-1 Zone Transition Event occurs in Equipment : 18
4-6-2 Host defines Limit Attributes : 19
4-6-3 Host queries equipment for current Limits: 19

4-7. Status Data Collection 20

4-7-1 Request Equipment Status Report: 20
4-7-2 Request Equipment Status Variable Namelist: 20

4-8. On-line Identification 21

4-8-1 Host Initiated: 21
4-8-2 Loopback Diagnostic: 21

4-9. Alarm Management 22

4-9-1 Enable/disable Alarms: 22
4-9-2 Upload Alarm Information: 22
4-9-3 Send Alarm Report: 22

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4-10. Remote Control 23

4-10-1 Remote Control: 23

4-11. Equipment Constants 24

4-11-1 Host Sends Equipment Constants: 24
4-11-2 Host Equipment Constants Request: 24
4-11-3 Host Equipment Constant Name Request: 24
4-11-4 Operator Changes Equipment Constant: 25

4-12. Process Program Management 26

4-12-1 Process Program Created, Edited, or Deleted by Operator: 26
4-12-2 Process Program Deletion by the Host: 26
4-12-3 Process Program Directory Request: 27
4-12-4 Process Program Upload: 27
4-12-5 Process Program Download: 30

4-13. Equipment Terminal Services 33

4-13-1 Host sends information to an equipment's display: 33
4-13-2 Host sends information to an equipment's display (Case : operator unrecognizes message): 34
4-13-3 Operator sends information to the host: 36

4-14. Error Message 37

4-14-1 Message Fault Due to Unrecognized Device ID: 37
4-14-2 Message Fault Due to Unrecognized Stream Type: 37
4-14-3 Message Fault Due to Unrecognized Function Type: 37
4-14-4 Message Fault Due to Illegal Data Format: 38
4-14-5 Communication Fault Due to Transaction Timer Timeout: 38
4-14-6 Message Fault Due to data too long: 38
4-14-7 Message Fault Due to Conversation Timeout: 39

4-15. Clock 40

4-15-1 Equipment Request Time: 40
4-15-2 Host Instruct Equipment to Set Time: 40
4-15-3 Host Requests Equipment's Current Time Value: 40

4-16. Spooling 41

4-16-1 Define Spool: 41
4-16-2 Define the Set of Messages to be Spooled: 42
4-16-3 Defines the Maximum Number of Messages to Send in Response to S6F23: 42
4-16-4 Request or Delete Spooled Data(MaxSpoolTransmit = 0): 43
4-16-5 Request or Delete Spooled Data(MaxSpoolTransmit > 0): 44

4-17. Control(Operator-Initiated Scenarios) 45

4-17-1 Host Accepts ON-LINE: 45
4-17-2 Host Denies ON-LINE: 45
4-17-3 Operator Sets OFF-LINE: 45
4-17-4 Operator Sets REMOTE: 46
4-17-5 Operator Sets LOCAL: 46

4-18. Control(Host-Initiated scenarios) 47

4-18-1 Host Sets OFF-LINE: 47
4-18-2 Host Sets ON-LINE: 48
4-18-3 Host Sets REMOTE: 48
4-18-4 Host Sets LOCAL: 49

5. SECS-II Message 50

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5-1. HOST Initiated Message 50

5-2. Equipment Initiated Message 51

6. Message Detail 52

6-1. Stream1 Equipment Status 52

6-2. Stream2 Equipment Control and Diagnostics 56

6-3. Stream5 Exception Handling 72

6-4. Stream6 Data Collection 75

6-5. Stream7 Process Program Management 79

6-6. Stream9 System Errors 85

6-7. Stream10 Terminal Services 87

7. Data Item Dictionary 89

7-1. Item Dictionary 89

7-2. Item Format Code 100

8. Setup 101

8-1. LAN cable 101

8-2. HSMS Setting 101

8-3. GEM Setting 101

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0. Documentation History

Revision	Date	History	Author
1.0	2016/09/26	First	K.Nishida
1.1	2017/06/29	Add the Exception of S2F35 Fix the following item format ALID、DATAID、DATALENGTH、ECID、 REPGSZ、RPTID、SMPLN、TRID、VID	K.Nishida
1.2	2017/07/07	・In 4-16 Spooling, Add that the spooling function is off by default. ・in 2-2 Equipment Processing, change state Pause to Abort. And Add the trigger #10	K.Nishida
1.3	2017/11/7	.2-2 Equipment Processing State Model Add the trigger “10”.	K.Nishida
1.4	2018/7/17	Behavior of S9F9 compliant with E5 standard	K.Nishida
1.5	2020/5/28	・Change page number to serial number. ・Corrected the function name and communication direction of S7F30	R.Nakazawa
1.6	2021/07/28	・Change S7F27 Message Format	Y.Ishikawa

1. General Requirements

This Host Communication System is based on below Standards

	Title	Version
SECS-II	SEMI Equipment Communications Standard 2, Message Content【SECS-II】	E5-0704
GEM	SEMI Generic Model for Communications and Control of SEMI Equipment【GEM】	E30-1103
HSMS	High-Speed SECS Message Services(HSMS) Generic Services.	E37-0303

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2. State Models

2-1 Communications and Control State Model

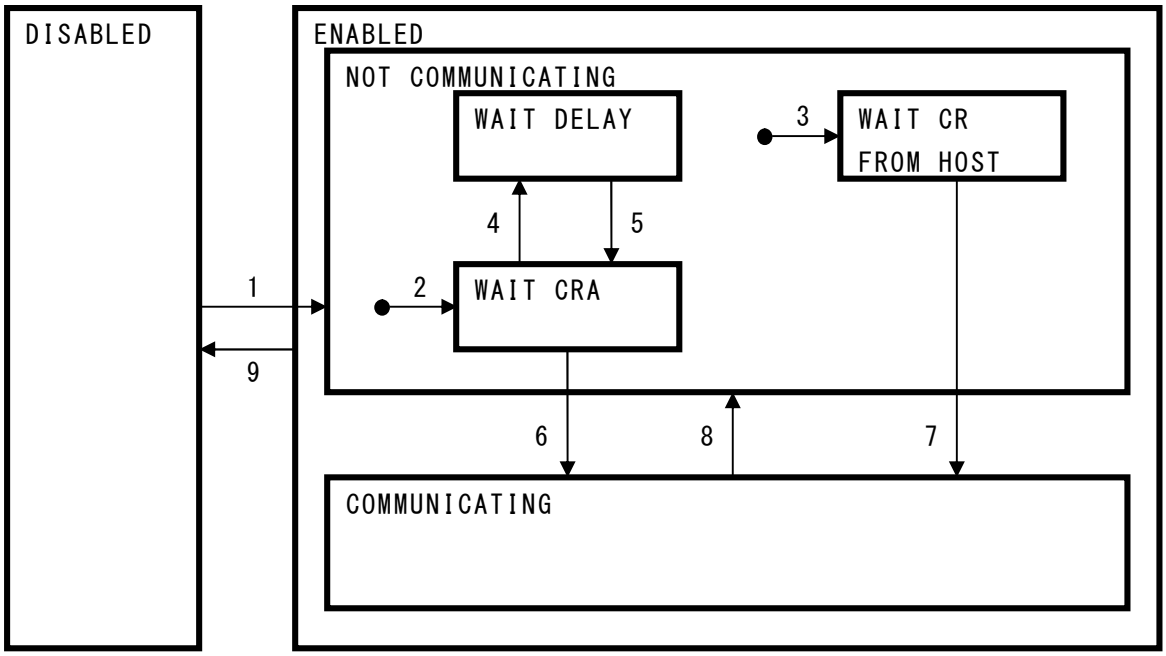
Communications and Control State Model is following.

DISABLED	: SECS-II Communication is disabled		
ENABLED	: SECS-II Communicatino is enabled		
	NOT COMMUNICATING	: The equipment is waiting Establish Communications. Establesh Communications is completed which the Host or the equipment send S1F13 and receive S1F14.	
		WAIT CRA	: Waiting an Establish Communications Acknowledge.
		WAIT DELAY	: Waiting between attempts to Establish Communications.
		WAIT CR FROMHOST	: Wait for the successful completion of the S1F14.
	COMMUNICATING	: Communications have been established.	
		OFF-LINE	: Impossible to transfer valid message.
			EQUIPMENT OFF-LINE : Waiting operator instructions to attempt on-line.
			ATTEMPT ON-LINE : Attempts to send an S1F1 to the host and wait reply.
			HOST OFF-LINE : Host changed to OFF-LINE
		ON-LINE	: SECS-II message may be exchanged and acted upon
			LOCAL : The equipment is operated only by operator.
			REMOTE : The host may operate the equipment

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2-1-1.Communication State Diagram

Default is NOT COMMUNICATING.



2-1-2.Communication State Table

Current State	#	Trigger	New State
DISABLED	1	Operator switches to enabled.	NOT COMMUNICATING
NOT COMMUNICATING	2	Default	WAIT CRA
	3	You can prohibite to change to WAIT CRA state(Option)	WAIT CR FROM HOST
WAIT CRA	4	Connection transaction failure.	WAIT DELAY
WAIT DELAY	5	CommDelay timer expired	WAIT CRA
WAIT CRA	6	Received expected S1F14 with COMMACK = 0	COMMUNICATING
WAIT CR FROM HOST	7	Received expected S1F13	COMMUNICATING
COMMUNICATING	8	Communication failure S9F9 command happened	NOT COMMUNICATING
ENABLE	9	Operator switches to DISABLED	DISABLED

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2-1-3.Control State Diagram

The default control state is Local when system start and the control state changed to ON-LINE from OFF-LINE.

※this default state is changed by S2F15.(ECV)

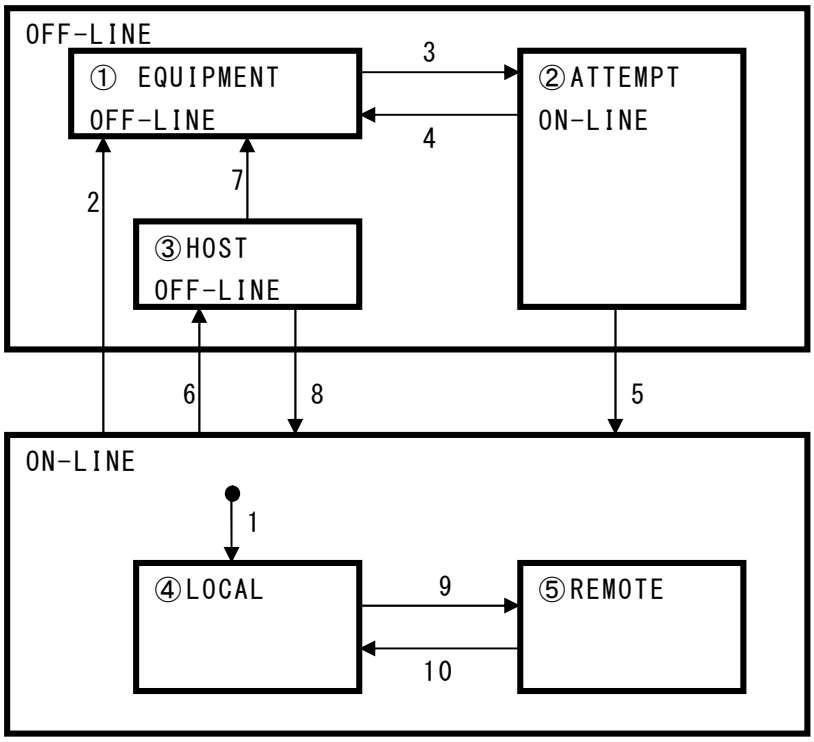
・InitControlState : The default control state when system start.

・InitOnlineState : The default control state the control state changed to ON-LINE from OFF-LINE

You can select the following state

・InitControlState(①～⑤)

・InitOnlineState(④⑤)



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2-1-4.Control State Table

Current State	#	Trigger	New State
	1	Default	LOCAL(you can select)
ON-LINE	2	Operator switches to OFF-LINE	EQUIPMENT OFF-LINE
EQUIPMENT OFF-LINE	3	Operator switches to ON-LINE HOST send S1F17(option)	ATTEMPT ON-LINE
ATTEMPT ON-LINE	4	Received S1F0 from HOST	EQUIPMENT OFF-LINE
ATTEMPT ON-LINE	5	Received S1F2 from HOST	ON-LINE
ON-LINE	6	Received S1F15 from HOST	HOST OFF-LINE
HOST OFF-LINE	7	Operator switches to OFF-LINE	EQUIPMENT OFF-LINE
HOST OFF-LINE	8	Received S1F17 from HOST	ON-LINE
LOCAL	9	Operator switches to REMOTE Received GO-REMOTE from HOST by S2F41.	REMOTE
REMOTE	10	Operator switches to LOCAL Received GO- LOCAL from HOST by S2F41	LOCAL

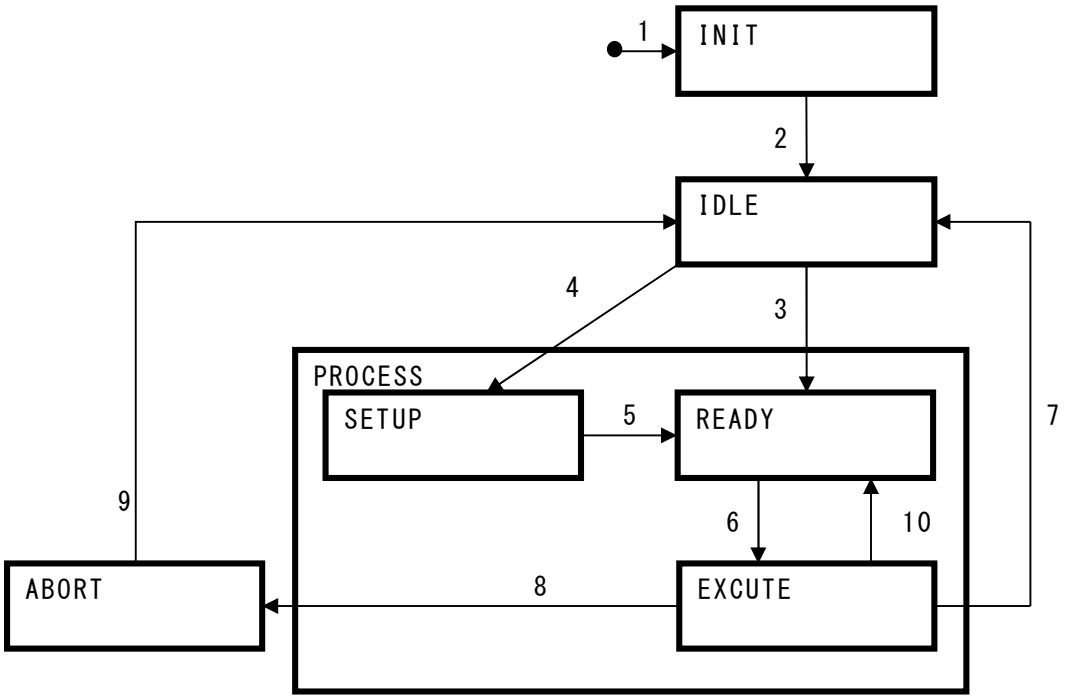
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2-2 Equipment Processing State Model

Equipment Processing State

INIT		The equipment is ready off or key switch off
IDLE		Initializing was completed
PROCESS		
	SETUP	Mantenace
	READY	The equipment is possible to process execution automatically
	EXCUTE	The equipment process execution automatically
	ABORT	The equipment happen some error and stoped.

Equipment Processing State Diaglam



Equipment Processing State

Current State	#	Trigger	New State
	1	Equipment power on	INIT
INIT	2	READY ON	IDLE
IDLE	3	Initialize completed	READY
IDLE	4	Maintenance start	SETUP
SETUP	5	Maintenance completed	READY
READY	6	Operator switchs to start	EXCUTE
EXCUTE	7	Process completed	IDLE
EXCUTE	8	Equipment happen errors	ABORT
ABORT	9	Operator release all errors	IDLE
EXCUTE	10	Operator switchs to stop	READY

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3. Gem Compliance Statement

GEM COMPLIANCE STATEMENT		
FUNDAMENTAL GEM QUIREMENTS	**IMPLEMENTED**	**GEM COMPLIANT**
State Models	Yes	Yes
Equipment Processing States	Yes	
Host-Initiated S1, F13/F14 Scenario	Yes	
Event Notification	Yes	
On-Line Identification	Yes	
Error Messages	Yes	
Documentation	Yes	
Control(Operator Initiated)	Yes	
ADDITIONAL CAPABILITIES	**IMPLEMENTED**	**GEM COMPLIANT**
Establish Communications	Yes	Yes
Dynamic Event Report Configuration	Yes	Yes
Variable Data Collection	Yes	Yes
Trace Data Collection	Yes	Yes
Status Data Collection	Yes	Yes
Alarm Management	Yes	Yes
Remote Control	Yes	No
Equipment Constants	Yes	No
Process Program Management	Yes	Yes
Material Movement	Yes	Yes
Equipment Terminal Services	Yes	Yes
Clock	Yes	Yes
Limits Monitoring	Yes	Yes
Spooling	Yes	Yes
Control(Host-Initiated)	Yes	Yes

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4. Gem Scenario

4-1. Establish Communications

You can establish communication between the Host and the equipment.
The equipment spool the data during discommunicating.
So, the host should request or delete the spooled data after establishing communication.
(refer 4-16-4 Request or Delete Spooled Data)

4-1-1 Host Attempts To Establish Communications

To send to when to start the communication.

COMMENT	HOST	EQUIPMENT
	Communications State = NOT COMMUNICATING (ENABLED)	
Establish Communications Request	S1F13→	
	←S1F14	Reply COMMACK=0 Communications st

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4-1-2 Equipment Attempts To Establish Communications and Host Acknowledges

To send to when to start the communication.

COMMENT	HOST	EQUIPMENT	COMMENT
	Communications State = NOT COMMUNICATING (ENABLED)		
		[LOOP①]	
		[LOOP②]	--SEND
	←S1F13		Establish Communications Request
Establish Communications Acknowledge	S1F14→		
		[IF]	S1F14 is recieved without timeouts
		[THEN]	Exit loop--LOOP②
		[ELSE]	Delay for interval in Establish Communications Timeout.
		[ENDIF]	
		[END LOOP②]	--SEND
		[IF]	COMMACK=0(Accept)
		[THEN]	Communications state =COMMUNICATING Exit loop —LOOP①
		[ELSE]	Reset timer for delay, and delay for interval specified in EstablishCommunications Timeout
		[ENDIF]	
		[END LOOP①]	

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4-1-3 Equipment Receives S1F14 From Host Before Sending S1F14

COMMENT	HOST	EQUIPMENT	COMMENT
	Communications State = NOT COMMUNICATING (ENABLED)		
		←S1F13	Establish Communications Request
Establish Communications Request	S1F13→		
Reply COMMACK=0(Accept)	S1F14→		S1F14 received from Host and
			Communications established and
			Communications state = COMMUNICATING
		←S1F14	Reply COMMACK=0(Accept)

4-1-4 Equipment Sends S1F14 To Host Before Receiving S1F14

COMMENT	HOST	EQUIPMENT	COMMENT
	Communications State = NOT COMMUNICATING (ENABLED)		
		←S1F13	Establish Communications Request
Establish Communications Request	S1F13→		
		←S1F14	Reply COMMACK=0(Accept)
			Communications established and
			Communications state = COMMUNICATING
Reply COMMACK=0(Accept)	S1F14→		S1F14 received from Host

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4-2. Event Notification

4-2-1 Collection Event which occurs on the Equipment

To notify the events on the device to the host.

However, the notification only enabled events.

Switching enable / disable events, the use of S2F37.

Event list see Attachment (Chapter2)

COMMENT	HOST	EQUIPMENT		COMMENT
			[IF]	Event Report is Multi-block
		←S6F5	[THEN]	send Multi-block inquire
Multi-block grant	S6F6 →			
			[ENDIF]	
		←S6F11		Equipment sends Event Report
Host acknowledges Event Report	S6F12→			

4-2-2 Host Requests Event Report

The host can require a series of reports linked to an event at a device by appointing CEID.

COMMENT	HOST	EQUIPMENT	COMMENT
Host requests an event report	S6F15→		
		←S6F16	Equipment sends event report

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4-3. Dynamic Event Report Configuration

4-3-1 Collection Event Reporting Set-up

The equipment follow the setting function.

Function	S/F
Define or delete report	S2F33
Link or unlink defined report to a event	S2F35
Enable or disable specific collection events	S2F37

※You can check these current setting on the EQPanel monitor(refer Chapter3)

COMMENT		HOST	EQUIPMENT	COMMENT
[IF]	Define Report is Multi-block			
[THEN]	send Multi-block inquire	S2F39→		
			←S2F40	Multi-block grant
[ENDIF]				
	Send report definitions	S2F33→		DATAIDs,RPTIDs and VIDs received
			←S2F34	DRACK=0 (the reports are OK)
[IF]	Link Event/Report if Multi-block			
[THEN]	send Multi-block inquire	S2F39→		
			←S2F40	Multi-block grant.
[ENDIF]				
	Link reports to events	S2F35→		CEIDs and the corresponding RPTIDs are received.
			←S2F36	LRACK=0 (the event linkages are Acceptable)
	Enable specific collection events	S2F37→		Enable/disable codes(CEEDs) and the respective event reporting CEIDs received.
			←S2F38	ERACK=0(OK)
				generate the specified reports when the appropriate collection events happen.

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4-4. Variable Data Collection

4-4-1 Host Requests Report

Host can request a report including the variable dat by specifying the only defined report ID.

COMMENT	HOST	EQUIPMENT	COMMENT
Host requests data variables contained in report RPTID	S6F19→		
		←S6F20	Equipment responds with a list of variable data for the given RPTID.

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4-5. Trace Data Collection

You can gether the regularly sampling data.

4-5-1 Host Initiates Trace Report :

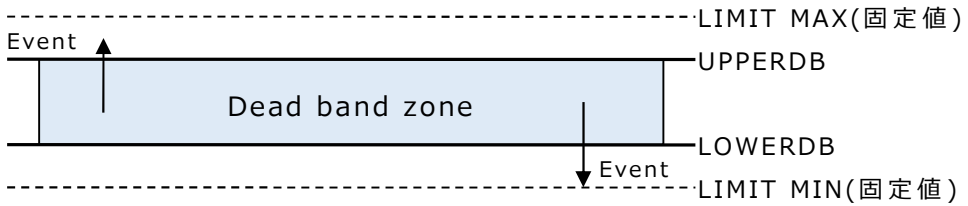
Trace condition setting and report the result from equipment.

COMMENT	HOST	EQUIPMENT	COMMENT
Trace Data initialization requested.	S2F23→		
		←S2F24	Acknowledge,trace initiated
			[DO①]repeat (Total sampling count(TOTSMP)
			÷report sampling count(REPGSZ))
			[DO②]repeat report sampling count(REPGSZ)
			Gather data SVID1,...,SVIDn
			every sampling period(DESPER).
			[END_DO②]
		←S6F1	Send SV1,...,SVn
Acknowledge receipt	S6F2 →		
			[END_DO①]
Request trace termination prior to completion(set the TOTSMP=0)	S2F23→		
		←S2F24	Acknowledge premature termination

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4-6. Limits Monitoring

You can monitor and check the value which can be adapt limit value.
Variable Limit Attributes (UPPERDB or LOWERDB) makes Deadband zone, the equipment occurs events at zone transition.



4-6-1 Zone Transition Event occurs in Equipment :
the equipment occurs events at zone transition

COMMENT	HOST	EQUIPMENT	COMMENT
			[IF] Event Report is Multi-block
		←S6F5	[THEN] send Multi-block inquire
Multi-block grant	S6F6 →		
			[ENDIF]
		←S6F11	Equipment sends Event Report
Host acknowledges Event Report	S6F12→		

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4-6-2 Host defines Limit Attributes :

Host can define the limit attributes for SV which can be adapt limit value

COMMENT	HOST	EQUIPMENT	COMMENT
[IF] S2F45 is Musti-block			
[THEN] Send Multi-block inquire	S2F39→		
		←S2F40	Multi-block grant
[END IF]			
Host defines new variable limit attributes	S2F45→		
		←S2F46	Equipment acknowledges host request.

4-6-3 Host queries equipment for current Limits:

Host can request the current limit attribute definitions.

COMMENT	HOST	EQUIPMENT	COMMENT
Host request equipment for current variable limit attribute definitions.	S2F47→		
		←S2F48	Equipment returns report containing requested variable limit attribute

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4-7. Status Data Collection

4-7-1 Request Equipment Status Report:

Host can request the status value by sending SVID

COMMENT	HOST	EQUIPMENT	COMMENT
Host requests report of selected status variable values.	S1F3→		
		←S1F4	Equipment responds report of selected status variable values

4-7-2 Request Equipment Status Variable Namelist:

Host can request the name list of the status values

COMMENT	HOST	EQUIPMENT	COMMENT
Host requests the name list of selected status variable values	S1F11→		
		←S1F12	Equipment responds the name list of selected status variable values.

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4-8. On-line Identification

4-8-1 Host Initiated:

Host can check the equipment is online.

COMMENT	HOST	EQUIPMENT	COMMENT
Are you there?	S1F1→		
		←S1F2	Equipment replies with MDLN and SOFTREV

4-8-2 Loopback Diagnostic:

Host can check the protocol and the communication circuit.

COMMENT	HOST	EQUIPMENT	COMMENT
Loopback Diagnostic Request	S2F25→		
		←S2F26	The binary string(ABS) sent is echoed back

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4-9. Alarm Management

4-9-1 Enable/disable Alarms:

The host can select the enable or disable of specific alarm by sending ALID

COMMENT	HOST	EQUIPMENT	COMMENT
Enable/disable Alarm	S5F3→		
		←S5F4	Acknowledge

4-9-2 Upload Alarm Information:

The host can get the alarm data and ALX of specific alarm by sending ALID..

COMMENT	HOST	EQUIPMENT	COMMENT
Request alarm data/text	S5F5→		
		←S5F6	Send alarm data/text

4-9-3 Send Alarm Report:

To notify the generated alarm to the host..

COMMENT	HOST	EQUIPMENT	COMMENT
		Alarm occurrence detected by the equipment.	
		←S5F1	Send alarm report(if enabled)
Acknowledge	S5F2→		
		←S6F11	Send event report(if enabled)
Acknowledge	S6F12→		

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4-10. Remote Control

4-10-1 Remote Control:

The host acts for the operation for the equipment by sending Host command.

Labor saving, prevention of erroneous operation can in this way come true

COMMENT	HOST	EQUIPMENT	COMMENT
Host Command Send	S2F41→		
		←S2F42	Host Command Acknowledge HCACK=0(assume command can be performed).
		←S6F11	Event Report-state change or other collection event occurrence.
Event Report Acknowledge	S6F12→		

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4-11. Equipment Constants

4-11-1 Host Sends Equipment Constants:

Host can change the one or more equipment constants (ECV) by sending ECID.

COMMENT	HOST	EQUIPMENT	COMMENT
Host sends equipment constants	S2F15→		
		←S2F16	EAC=0 equipment sets constants

4-11-2 Host Equipment Constants Request:

Host can request the one or more equipment constants (ECV) by sending ECID.

COMMENT	HOST	EQUIPMENT	COMMENT
Host requests equipment constants	S2F13→		
		←S2F14	Equipment constant data

4-11-3 Host Equipment Constant Name Request:

Host can request the one or more equipment constants (ECV) name by sending ECID。

COMMENT	HOST	EQUIPMENT	COMMENT
Host request constant names	S2F29→		
		←S2F30	Equipment constant names

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4-11-4 Operator Changes Equipment Constant:

If the operator changes the equipment constant, and notifies the event to the host.

COMMENT	HOST	EQUIPMENT	COMMENT
			Operator changes equipment constant at equipment operator console
		←S6F11	Equipment reports equipment constant change
Host acknowledges event	S6F12→		

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4-12. Process Program Management

4-12-1 Process Program Created, Edited, or Deleted by Operator:

COMMENT	HOST	EQUIPMENT	COMMENT
			New process program created, edited, or deleted by operator at equipment.
			PPChangeName = PPID
			PPChangeStatus = 1 (Created)
			= 2 (Edited)
			= 3 (Deleted)
			[IF] CEID for Process Program Change Event (PPCHANGE_EVENT) is enabled
			[THEN]
		←S6F11	Send Event Report
Event Report Acknowledge	S6F12→		

4-12-2 Process Program Deletion by the Host:

To delete a process program that is registered to the equipment

COMMENT	HOST	EQUIPMENT	COMMENT
Delete Process Program Send	S7F17→		
		←S7F18	Delete Process Program Acknowledge.
			The process program is removed from non-volatile storage..

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4-12-3 Process Program Directory Request:

Host can request the all name list of process program registered in equipment

COMMENT	HOST	EQUIPMENT	COMMENT
Current EPPD Request	S7F19→		
		←S7F20	Current EPPD Data (the all name list of process program registered in equipment)

4-12-4 Process Program Upload:

Host-Initiated formatted Process Program Upload :

Host can request the formatted process program upload.

COMMENT	HOST	EQUIPMENT	COMMENT
Host request to upload the formatted Process program	S7F25→		
		←S7F26	Send the formatted Process program

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Equipment-Initiated formatted Process Program Upload :

Operators can send the formatted Process Program to Host

COMMENT	HOST	EQUIPMENT	COMMENT
			[IF] Process program is multi-block
			[THEN]
		←S7F1	Process Program Load Inquire
Process Program Load Grant	S7F2→		
			[END_IF]
			(You can omit the process program load inquire scenario)
			(Option)
		←S7F23	Send the formatted Process program
Formatted Process Program Acknowledge	S7F24→		

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Host-Initiated unformatted Process Program Upload :

Host can request the unformatted process program upload.

COMMENT	HOST	EQUIPMENT	COMMENT
Host request to upload the unformatted Process program	S7F5 →		
		←S7F6	Send the unformatted Process program

Equipment-Initiated unformatted Process Program Upload :

Operators can send the unformatted Process Program to Host

COMMENT	HOST	EQUIPMENT	COMMENT
			[IF] Process program is multi-block
			[THEN]
		←S7F1	Process Program Load Inquire
Process Program Load Grant	S7F2→		
			[END_IF]
			(You can omit the process program load inquire scenario)
			(Option)
		←S7F3	Send the unformatted Process program
Unformatted Process Program Acknowledge	S7F4→		

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4-12-5 Process Program Download:

Host-Initiated Formatted Process Program Download :

COMMENT	HOST	EQUIPMENT	COMMENT
[IF] Process program is multi-block			
[THEN]			
Process Program Load Inquire	S7F1→		
		←S7F2	Process Program Load Grant
[END_IF]			
(You can omit the process program load inquire scenario)(Option)			
Send the formatted Process program	S7F23→		
		←S7F24	formatted Process Program Acknowledge
			[IF]S7F27 is multi-block
			[THEN]
		←S7F29	Send Process Program Verification Inquire
Reply Process Program Verification Grant	S7F30→		
			[END_IF]
		←S7F27	Process Program Verification Send
Process Program Verification Acknowledge	S7F28→		

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Operators can send to request the formatted Process Program to Host

COMMENT	HOST	EQUIPMENT	COMMENT
		←S7F25	Equipment request to download the formatted Process program
Send the formatted Process program	S7F26→		
			[IF]S7F27 is multi-block
			[THEN]
		←S7F29	Send Process Program Verification Inquire
Reply Process Program Verification Grant	S7F30→		
			[END_IF]
		S7F27	Process Program Verification Send
Process Program Verification Acknowledge	S7F28→		

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Host-Initiated Unformatted Process Program Download :

COMMENT	HOST	EQUIPMENT	COMMENT
[IF] Process program is multi-block			
[THEN]			
Process Program Load Inquire	S7F1→		
		←S7F2	Process Program Load Grant
[END_IF]			
(You can omit the process program load inquire scenario)(Option)			
Send the Unformatted Process program	S7F3→		
		←S7F4	Unformatted Process Program Acknowledge

Operators can send to request the Unformatted Process Program to Host

COMMENT	HOST	EQUIPMENT	COMMENT
		←S7F5	Equipment request to download the Unformatted Process program
Send the Unformatted Process program	S7F6→		

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4-13. Equipment Terminal Services

4-13-1 Host sends information to an equipment's display:

Host can send the text. The text is displayed by popup message.

COMMENT	HOST	EQUIPMENT	COMMENT
Host sends textual information to equipment for display to the operator on terminal 0.	S10F3→		
		←S10F4	Equipment acknowledges request to display text. (Equipment show received message by popup) Operator indecates message recognition (Operator clear the popup)
		←S6F11	Message recognition event (TerminalConfirmed)
Host acknowledges	S6F12→		
		←S10F1	Operator responds with text via terminal 0
Host acknowledges receipt of operator text	S10F2→		

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4-13-2 Host sends information to an equipment's display (Case : operator unrecognizes message):

Host can send the text. The text is displayed by popup message.

COMMENT	HOST	EQUIPMENT	COMMENT
Host sends textual information to equipment for display to the operator on terminal 1	S10F3→		
		←S10F4	Equipment acknowledges request to display text. (Equipment show received message by popup)
Host sends textual information to equipment for display to the operator on terminal 1, because operator unrecognized first message.	S10F3→		
		←S10F4	Equipment acknowledges request to display text. (Equipment show received message by popup And the first message is not cleared.) Operator indecates message recognition (Operator clear the popup)
		←S6F11	Message recognition event (TerminalConfirmed)
Host acknowledges	S6F12→		

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Host can send the multi block text. The text is displayed by popup message

COMMENT	HOST	EQUIPMENT	COMMENT
Host sends textual information to equipment for display to the operator on terminal 1	S10F5→		
		←S10F6	Equipment acknowledges request to display text. (Equipment show received message by popup And the first message is not cleared.) Operator indecates message recognition (Operator clear the popup)
		←S6F11	Message recognition event (TerminalConfirmed)
Host acknowledges	S6F12→		

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4-13-3 Operator sends information to the host:

Operator can send the text message.

COMMENT	HOST	EQUIPMENT	COMMENT
		←S10F1	Operator sends textual information via equipment terminal x
Host acknowledges receipt of operator initiated message.	S10F2→		
Host responds with information for display to the operator on terminal x	S10F3→		
		←S10F4	Equipment acknowledges request to display text. (Equipment show received message by popup And the first message is not cleared.) Operator indecates message recognition (Operator clear the popup)
		←S6F11	Message recognition event (TerminalConfirmed)
Host acknowledges	S6F12→		

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4-14. Error Message

4-14-1 Message Fault Due to Unrecognized Device ID:

COMMENT	HOST	EQUIPMENT	COMMENT
Host sends a message.	SxFy→		
			Equipment detects an unrecognized device ID within the message from the host
		←S9F1	Equipment reports to the host that an "unrecognized device ID" was detected in the received message.

4-14-2 Message Fault Due to Unrecognized Stream Type:

COMMENT	HOST	EQUIPMENT	COMMENT
Host sends a message.	SxFy→		
			Equipment detects an unrecognized stream type within the message from the host.
		←S9F3	Equipment reports to the host that an "unrecognized stream type" was detected in the received message.

4-14-3 Message Fault Due to Unrecognized Function Type:

COMMENT	HOST	EQUIPMENT	COMMENT
Host sends a message.	SxFy→		
			Equipment detects an unrecognized function type within the message from the host.
		←S9F5	Equipment reports to the host that an "unrecognized function type" was detected in the received message.

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4-14-4 Message Fault Due to Illegal Data Format:

COMMENT	HOST	EQUIPMENT	COMMENT
Host sends a message.	SxFy→		
			Equipment detects illegal data format within the message from the host
		←S9F7	Equipment reports to the host that "illegal data format" was detected in the received message.

4-14-5 Communication Fault Due to Transaction Timer Timeout:

The communication state change to NOT COMMUNICATING after this message send, because system keep appropriate state

COMMENT	HOST	EQUIPMENT	COMMENT
			Equipment does not receive an expected reply message from the host and a transaction timer timeout occurs.
		←S9F9	Equipment reports to the host that a transaction timer timeout occurred

4-14-6 Message Fault Due to data too long:

COMMENT	HOST	EQUIPMENT	COMMENT
Host sends a message.	SxFy→		
			Equipment detects that the message from the host contains more data than it can handle..
		←S9F11	Equipment reports to the host that "data too long" was detected in the received message.

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4-14-7 Message Fault Due to Conversation Timeout:

The specific message don't send in the setting time

COMMENT	HOST	EQUIPMENT	COMMENT
Host sends a message.	S7F1→		
			Though data should have been received, Host was not able to receive it before appropriate time.
		←S9F13	Equipment reports to the host that a conversation timeout.

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4-15. Clock

This scenario is used for a time-base check of equipment and synchronization with host time-base.

4-15-1 Equipment Request Time:

COMMENT	HOST	EQUIPMENT	COMMENT
		←S2F17	Equipment requests a time value from the host
Host responds with a TIME value	S2F18→		
			Equipment sets its internal time reference to the value of TIME received from the host.

4-15-2 Host Instruct Equipment to Set Time:

COMMENT	HOST	EQUIPMENT	COMMENT
Host instructs equipment to set its time	S2F31→		
			Equipment sets its internal time reference to the value of TIME received from the host
		←S2F32	Send acknowledges

4-15-3 Host Requests Equipment's Current Time Value:

COMMENT	HOST	EQUIPMENT	COMMENT
Host requests equipment time	S2F17→		
		←S2F18	Equipment returns its internal time

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4-16. Spooling

The Spooling function is effective only when control status is “ON-LINE”.

This function is off by default.

The maximum size of spooled data is 1M byte.

4-16-1 Define Spool:

Host can change the Spool enable or disable by setting equipment const

However, Host should request or delete spooling data if equipment already spool some data.

COMMENT	HOST	EQUIPMENT		COMMENT
Send the equipment constant(EnableSpooling)	S2F15→			
		←S2F16	Equipment acknowledges equipment constant change	

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4-16-2 Define the Set of Messages to be Spooled:

This Scenario is used to set up the list of messages that the equipment should spool (or by defining none, to disable spooling).This setting is necessary before spooling start (before not communicating)

COMMENT	HOST	EQUIPMENT	COMMENT
Host defines messages to be spooled in case of communications failure.	S2F43→		
		←S2F44	Equipment acknowledges setup

The message list of possible spooling
S2F17
S5F1
S6F1
S6F11
S7F3
S7F5
S10F1

4-16-3 Defines the Maximum Number of Messages to Send in Response to S6F23:

Host can request the spooled data by S6F23, and set the spooled data count which the equipment send one time by setting equipment constant data (ECV: MaxSoolTransmit).

When MaxSoolTransmit = 0, the equipment send all spooled data one time.

COMMENT	HOST	EQUIPMENT	COMMENT
Host sends value for equipment constant MaxSpoolTransmit.	S2F15→		
		←S2F16	Equipment acknowledges equipment constant change

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4-16-4 Request or Delete Spooled Data(MaxSpoolTransmit = 0):

This Scenario is used to initiate the transfer of the spooled data from the equipment to the host or to purge the spool

COMMENT	HOST	EQUIPMENT	COMMENT
	Communications were lost and then re-established		
Host requests data that includes spool related status variables. NOTE:S1F3 is one of various methods that could be used	S1F3→		
		←S1F4	Send status data (Ex,SpoolCountActual = 8, MaxSpoolTransmit = 0).
Request or delete spooled data	S6F23→		
		←S6F24	Request spooled data acknowledgement. [IF]RSDC = 0(Spool data requested.) [THEN] The appropriate Streams and Functions are used to transmit the spooled data to the host. [ELSE_IF]RSDC = 1 (Spool data deleted) [THEN] Spool data discarded. [END_IF]
		←S6F11	Send spooling activated event (SpoolingActived)
Acknowledge	S6F12→		
		←SxFx	Send the all spooled data.
Acknowledge	SxFx+1→		
		←S6F11	Send spooling deactivated event (SpoolingDeactived)
Acknowledge	S6F12→		

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4-16-5 Request or Delete Spooled Data(MaxSpoolTransmit > 0):

This Scenario shows the affect of MaxSpoolTransmit<SpoolCountActual on the Spool Transmit process. For the purpose of illustration, the value of MaxSpoolTransmit is 5 and the SpoolCountActual is 8 (at the time communications are re-established).No messages are added to the Spool during the transmit process.

COMMENT	HOST	EQUIPMENT	COMMENT
	通信が中断し,その後もう一度確立された		
Host requests data that includes spool related status variables.	S1F3→		
		←S1F4	Send status data(e.g.SpoolCountActual = 8, MaxSpoolTransmit = 5).
Host requests spooled data(RSDC=0)	S6F23→		
		←S6F24	Request spooled data acknowledgement.
		←S6F11	Send spooling activated event (SpoolingActived)
Acknowledge	S6F12→		
		←SxFx	The five oldest messages in the Spool are transmitted to the host.
Acknowledge	SxFx+1→		
Host recognizes that MaxSpoolTransmit is reached. Host requests additional spooled data (RSDC=0).	S6F23→		
		←S6F24	Request spooled data acknowledgement
		←SxFx	The three remaining messages are transmitted from the spool.
Acknowledge	SxFx+1→		
		←S6F11	Send spooling deactivated event (SpoolingDeactived)
Acknowledge	S6F12→		

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4-17. Control(Operator-Initiated Scenarios)

4-17-1 Host Accepts ON-LINE:

Operators change to ON-LINE.

COMMENT	HOST	EQUIPMENT	COMMENT
		Operator switch to ON-LINE when communication state is equipment OFF-LINE.	
		←S1F1 Equipment requests ON-LINE	
Host grants ON-LINE	S1F2→		
		←S6F11 "Control State LOCAL(or REMOTE)"event happen	
Acknowledge	S6F12→		

4-17-2 Host Denies ON-LINE:

Communicating can not be started if Host send S1F0.

COMMENT	HOST	EQUIPMENT	COMMENT
		Operator switch to ON-LINE when communication state is equipment OFF-LINE.	
		←S1F1 Equipment requests ON-LINE	
Host denies ON-LINE	S1F0→		

4-17-3 Operator Sets OFF-LINE:

Operators change to OFF-LINE.

COMMENT	HOST	EQUIPMENT	COMMENT
		Operator switch to OFF-LINE when communication state is equipment ON-LINE.	
		←S6F11 "OFF-LINE" event happen	
Acknowledge	S6F12→		

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4-17-4 Operator Sets REMOTE:

Operator change to Remote state.

COMMENT	HOST	EQUIPMENT	COMMENT
		Operator switch to REMOTE.	
		←S6F11 “Control State :Remote” event happen	
Acknowledge	S6F12→		

4-17-5 Operator Sets LOCAL:

Operator change to Local state.

COMMENT	HOST	EQUIPMENT	COMMENT
		Operator switch to LOCAL.	
		←S6F11 “Control State :Local” event happen	
Acknowledge	S6F12→		

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4-18. Control(Host-Initiated scenarios)

4-18-1 Host Sets OFF-LINE:

COMMENT	HOST	EQUIPMENT		COMMENT
Host requests OFF-LINE.	S1F15→			
			[IF] Equipment is OFF-LINE	
			[THEN]	
		←S1F0	Equipment denies requests	
			[ELSE] Equipment is ON-LINE	
		←S1F16	Equipment acknowledges request and change to OFF-LINE	
		←S6F11	"Equipment OFF-LINE"event.happen	
Acknowledge	S6F12→			
			[END_IF]	

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4-18-2 Host Sets ON-LINE:

COMMENT	HOST	EQUIPMENT		COMMENT
Host requests ON-LINE.	S1F17→			
			[IF]	HOST OFF-LINE state and DISABLED.
			[THEN]	
		←S1F18		Equipment denies request (ONLACK=1).
			[ELSE]	HOST ON-LINE
		←S1F18		Equipment denies request (ONLACK=2 ALREADY).
			[ELSE]	OFF-LINE and ENALBED
		←S1F18		Equipment acknowledges Request (ONLACK =0)
		←S6F11	"Control stat LOCAL(or REMOTE)"event happen	
Acknowledge	S6F12→			
			[END_IF]	

4-18-3 Host Sets REMOTE:

COMMENT	HOST	EQUIPMENT		COMMENT
Send Host command (GO-REMOTE)	S2F41→			
		S2F42	Send Host Command Acknowledge HCACK=0 after changed	
		←S6F11	"Control stat REMOTE"event happen	
Acknowledge	S6F12→			

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4-18-4 Host Sets LOCAL:

COMMENT	HOST	EQUIPMENT	COMMENT
Send Host command (GO-LOCAL)	S2F41→		
		S2F42	Send Host Command Acknowledge HCACK=0 after changed
		←S6F11	"Control stat LOCAL"event happen
Acknowledge	S6F12→		

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5. SECS-II Message

5-1. HOST Initiated Message

HOST INITIATES			EQUIPMENT RESPONSE		
S/F	NAME	MN	S/F	NAME	MN
S1Fx	Stream 1 messages		S1F0	Abort Transaction	S1F0
S1F1	Are You There Request	R	S1F2	On Line Data	D
S1F3	Selected Equipment Status Request	SSR	S1F4	Selected Equipment Status Request	SSD
S1F11	Status Variable Namelist Request	SVNR	S1F12	Status Variable Namelist Reply	SVNRR
S1F13	Establish Communications Request	CR	S1F14	Establish Communications Request	CRA
S1F15	Request OFF-LINE	ROFL	S1F16	OFF-LINE Acknowledge	OFLA
S1F17	Request ON-LINE	RONL	S1F18	ON-LINE Acknowledge	ONLA
S2Fx	Stream 2 messages		S2F0	Abort Transaction	S2F0
S2F13	Equipment Constant Request	ECR	S2F14	Equipment Constant Data	ECD
S2F15	New Equipment Constant Send	ECS	S2F16	New Equipment Constant	ECA
S2F17	Data and Time Request	DTR	S2F18	Data and time Data	DTD
S2F23	Trace Initialize Send	TIS	S2F24	Trace Initialize Acknowledge	TIA
S2F25	Loopback Diagnostic Request	LDR	S2F26	Loopback Diagnostic Data	LDD
S2F29	Equipment Constant Namelist	ECNR	S2F30	Equipment Constant Namelist	ECN
S2F31	Date and Time Set Request	DTS	S2F32	Date and Time Set Acknowledge	DTA
S2F33	Define Report	DR	S2F34	Define Report Acknowledge	DRA
S2F35	Link Event Report	LER	S2F36	Link Event Report Acknowledge	LERA
S2F37	Enable/Disable Event Report	EDER	S2F38	Enable/Disable Event Report	EERA
S2F39	Multi-block Inquire	DMBI	S2F40	Multi-block Grant	DMBG
S2F41	Host command Send	HCS	S2F42	Host Command Acknowledge	HCA
S2F43	Reset Spooling Streams and	RSSF	S2F44	Reset Spooling Acknowledge	RSA
S2F45	Define Variable Limit Attributes	DVLA	S2F46	Variable Limit Attribute	VLAA
S2F47	Variable limit Attribute Request	VLAR	S2F48	Variable Limit Attributes Send	VLAS
S5Fx	Stream 5 messages		S5F0	Abort Transaction	S5F0
S5F3	Enable/Disable Alarm Send	EAS	S5F4	Enable/Disable Alarm Acknowledge	EAA
S5F5	List Alarms Request	LAR	S5F6	List Alarm Data	LAD
S6Fx	Stream 6 messages		S6F0	Abort Transaction	S6F0
S6F15	Event Report Request	ERR	S6F16	Event Report Data	ERD
S6F19	Individual Report Request	IRR	S6F20	Individual Report Data	IRD
S6F23	Request Spooled Data	RSD	S6F24	Request Spooled Data	RSDAS
S7Fx	Stream 7 messages		S7F0	Abort Transaction	S7F0
S7F1	Process Program Load Inquire	PPI	S7F2	Process Program Load Grant	PPG
S7F3	Process Program Send	PPS	S7F4	Process Program Ack	PPA
S7F5	Process Program Request	PPR	S7F6	Process Program Data	PPD
S7F17	Delete Process Program Send	DPS	S7F18	Delete Process Program	DPA
S7F19	Current EPPD Request	RER	S7F20	Current EPPD Data	RED
S7F23	Formatted Process Program Send	FPS	S7F24	Formatted Process Program Ack	FPA
S7F25	Formatted Process Program Request	FPR	S7F26	Formatted Process Program Data	FPD
S10Fx	Stream 10 messages		S10F0	Abort Transaction	S10F0
S10F3	Terminal Display, Single	VTN	S10F4	Terminal Display, Single Ack	VTA
S10F5	Terminal Display, Single	VTN	S10F6	Terminal Display, Single Ack	VTA

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5-2. Equipment Initiated Message

EQUIPMENT INITIATES			HOST RESPONSE		
S/F	NAME	MN	S/F	NAME	MN
S1Fx	Stream 1 messages		S1F0	Abort Transaction	S1F0
S1F1	Are You There Request	R	S1F2	On Line Data	D
S1F13	Establish Communications Request	CR	S1F14	Establish Communications Request	CRA
S2Fx	Stream 2 message		S2F0	Abort Transaction	S2F0
S2F17	Data and Time Request	DTR	S2F18	Data and time Data	DTD
S5Fx	Stream 5 messages		S5F0	Abort Transaction	S5F0
S5F1	Alarm Report Send	ARS	S5F2	Alarm Report Acknowledge	ARA
S6Fx	Stream 6 messages		S6F0	Abort Transaction	S6F0
S6F1	Trace Data Send	TDS	S6F2	Trace Data Acknowledge	TDA
S6F5	Multi-Bloc Data Send Inquire	MBI	S6F6	Multi-Block Grant	MBG
S6F11	Event Report Send	ERS	S6F12	Event Report Acknowledge	ERA
S7Fx	Stream 7 messages		S7F0	Abort Transaction	S7F0
S7F1	Process Program Load Inquire	PPI	S7F2	Process Program Load Grant	PPG
S7F3	Process Program Send	PPS	S7F4	Process Program Ack	PPA
S7F5	Process Program Request	PPR	S7F6	Process Program Data	PPD
S7F23	Formatted Process Program Send	FPS	S7F24	Formatted Process Program Ack	FPA
S7F25	Formatted Process Program	FPR	S7F26	Formatted Process Program Data	FPD
S7F27	Process Program Verification Send	PVS	S7F28	Process Program Verification Ack	PVA
S7F29	Process Program Verification	PVI	S7F30	Process Program Verification Grant	PVG
S9F1	Unrecognized R-bit and Device ID	UDN	NONE		
S9F3	Unrecognized W-bit and Stream	USN	NONE		
S9F5	Unrecognized Function Type	UFN	NONE		
S9F7	Illegal Data	IDN	NONE		
S9F9	Transaction Timer Timeout	TTN	NONE		
S9F11	Data Too Long	DLN	NONE		
S9F13	Conversation Timeout	CTN	NONE		
S10Fx	Stream 10 messages		S10F0	Abort Transaction	S10F
S10F1	Terminal Request	TRN	S10F2	Terminal Request Acknowledge	TRA

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6 . Message Detail

6-1. Stream1 Equipment Status

S1F0 Abort Transaction(S1F0) S,H<->E

Description : Used is lieu of an expected replay to abort a transaction. Function 0 is defined in every stream and has the same meaning in every stream.

Structure: Header only

S1F1 Are You There Request(R) S,H<->E,reply

Description: Establishes if the equipment is on line. A function 0 response to this message means the communication is inoperative. In the equipment, a function 0 is equipment to a timeout on the receive timer after issuing S1F1 to the host.

Structure: Header only

S1F2 On Line Data(D) S,H<->E

Description: Data signifying that the equipment is alive

Structure: L,2
1.<MDLN>
2.<SOFTREV>

Exception: The host sends a zero-length list to the equipment.

Data Item : MDLN Equipment Model Type
SOFTREV Software revision code

S1F3 Selected Equipment Status Request(SSR) S,H->E,reply

Description: Host request the selected status value

Structure: The following structure is approved for all item formats and should be used by all new implementations.

L,n
1.<SVID1>
.
.
n.<SVIDn>

Exception: A zero-length list (n=0) means report all SVIDs
Data Item : SVID Status variable ID

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S1F4 Selected Equipment Status Data(SSD) M,H<-E

Description: The equipment reports the value of each SVID in the order requested. The host remembers the requested SVIDs.

Structure:
```
L,n
  1.<SV1>
  .
  .
  n.<SVn>
```

Exception: If n=0, no response can be made. A zero length returned for SVi means that SVIDi does not exist.

Data Item : SV Status variable data

S1F11 Status Variable Namelist Request(SVNR) S,H->E,reply

Description: Host can request the name list of specific SVIDs.

Structure:
```
L,n
  1.<SVID1>
  .
  .
  n.<SVIDn>
```

Exception: A zero length (n=0) means report all SVIDs

Data Item : SVID Status variable ID

S1F12 Status Variable Namelist reply(SVNRR) M,H<-E

Description: The equipment reports the name list of specific SVIDs

Structure:
```
L,n
  1.L,3
    1.<SVID1>
    2.<SVNAME1>
    3.<UNITS1>
  2.L,3
  .
  .
  n.L,3
    1.<SVIDn>
    2.<SVNAMEn>
    3.<UNITSn>
```

Data Item : SVID Status variable ID
SVNAME Status variable data
UNIT Unit and Notes

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S1F13 Establish Communications Request(CR) S,H<->E,reply

Description: The purpose of this message is to provide a formal means of initializing communications at a logical application level both on power-up and following a break in communications. It should be the first message sent following any period where host and Equipment SECS applications are unable to communicate. An attempt to send an Establish Communications Request(S1F13) should be repeated at programmable intervals until an Establish Communications Acknowledge(S1F14) is received within the transaction timeout period with an acknowledgement code accepting the establishment.

Structure: L,2
1.<MDLN>
2.<SOFTREV>

Exception: The host sends a zero-length list to the equipment
Data Item : MDLN Equipment Model Type
SOFTREV Software revision code

S1F14 Establish Communications Request Acknowledge(CRA) S,H<->E

Description: Accept or deny Establish Communications Request (S1F13).MDLN and SOFTREV are on-line data and are valid only if COMMAK = 0.

Structure: L,2
1. <COMMACK>
2. L,2
1.<MDLN>
2.<SOFTREV>

Exception: Host sends a zero –length list of item2 to the equipment
Data Item : COMMACK Establish Communications Acknowledge Code, 1byte
0 = Accepted
1 = Denied, Try Again
65 = System Error
MDLN Equipment Model Type
SOFTREV Software revision code

S1F15 Request OFF-LINE(ROFL) S,H->E,reply

Description: The host requests that the equipment transition to the OFF-LINE state.

Structure: Header only

S1F16 OFF-LINE Acknowledge(OFLA) S,H<-E

Description: Acknowledge or error of S1F15

Structure: <OFLACK>.
Data Item : OFLACK Acknowledge code for OFF-LINE request
0 = OFF-LINE Accepted
65 = System Error

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S1F17 Request ON-LINE(RONL) S,H->E,reply

Description: The host requests that the equipment transition to the ON-LINE state
The host can change to ON-LINE at equipment OFF-LINE.

Structure: Header only

S1F18 ON-LINE Acknowledge(ONLA) S,H<-E

Description: Acknowledge or error of S1F17

Structure: <ONLACK>
Data Item : ONLACK 0 = ON-LINE Accepted
1 = Not grant
2 = Already ON-LINE
65 = System Error

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6-2. Stream2 Equipment Control and Diagnostics

S2F0 Abort Transaction(S2F0) S,H<->E

Description: Same from as S1F0

S2F13 Equipment Constant Request(ECR) S,H ->E,reply

Description: Host can request the equipment constants (ECV)

Structure:
```
L,n
  1.<ECID1>
  .
  .
  n.<ECIDn>
```

Exception: A zero-length list (n=0) means report all ECV's according to a predefined order.

Data Item : ECID Equipment Constant ID

S2F14 Equipment Constant Data(ECD) M,H<-E

Description: Data Response to S2F13 in the order requested.

Structure:
```
L,n
  1.<ECV1>
  .
  .
  n.<ECVn>
```

Exception: A zero length ECVi means that ECIDi does not exist.

Data Item : ECV Equipment Constant Value

S2F15 New Equipment Constant Send(ECS) S,H->E,reply

Description: Change one or more equipment constant values

Structure:
```
L,n
  1.L,2
    1.<ECID1>
    2.<ECV1>
  2.L,2
  .
  .
  n.L,2
    1.<ECIDn>
    2.<ECVn>
```

Data Item : ECID Equipment Constant ID
ECV Equipment Constant Value

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S2F16 New Equipment Constant Acknowledge(ECA)	S,H<-E

Description: Acknowledge or error of S2F15. If EAC contains an error code (non-zero), the equipment should not change any of the ECIDs specified in S2F15.

Structure: <EAC>

Data Item : EAC Equipment acknowledge code, 1byte
0 = Acknowledge
1 = Denied, At least one ECID does not exis
2 = Denied, Busy
3 = Denied, At least one ECV is out of range
65 = System Error

S2F17 Date and Time request(DTR)	S,H<->E,reply

Description: Useful to check equipment time base or to synchronize with the host time base for equipment.

Structure: Header only

S2F18 Date and Time data(DTD)	S,H<->E

Description: Actual time data

Structure: <TIME>

Exception: The zero length item means the equipment dose not have clock.

Data Item : TIME The length is different depending on equipment constant (ECV:TimeFormat)

TimeFormat	Length	Format
0	12byte	YYMMDDhhmmss
1	16byte	YYYYMMDDhhmmsscc
2	14byte	YYYYMMDDhhmmss

YYYY = Year 0000 ～9999
MM = Month 01 ～12
DD = Day 01 ～31
hh = Hour 00 ～59
mm = Minuites 00 ～59
ss = Second 00 ～59
cc = 1/100 sec 00 ～99

note: This is format and is not precision

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S2F23 Trace Initialize Send(TIS) S,H->E,reply

Description: Host can set the way to trace SVs
10 traces can be carried out at the same time.
If Host select the same TRID as current carried out one, that TRID trace is stoped and restarted by new parameter.
If Host select the same TRID as current carried out one and set TOTSMP=0 or REPGSZ=0, that TRID trace is stopped.
When S2F23 is multi block, the equipment need S2F39/F40 before S2F23.
Don't set the overspec sampling period.

The equipment send the result of trace by S6F1(with TRID).

Structure: The following structure is approved for all item formats and should be used by all new implementations

```
L,2
  1.<TRID>
  2.<DSPER>
  3.<TOTSMP>
  4.<REPGSZ>
  5.L,n
    1,<SVID1>
    .
    .
    n<SVIDn>
```

Note: The maximum of same time trace is 10
The maximum of "n"(SVIDs) is 100

Data Item :

TRID	Trace ID
DSPER	Sampling period Minimum is 1second Hhmmss 6byte
TOTSMP	Total sampling count
REPGSZ	Report count every S6F1 When the equipment get the data this setting count, S6F1 is sent.
SVID	Status value ID

S2F24 Trace Initialize Acknowledge(TIA) S,H<-E

Description: Acknowledge or error of S2F24

Structure: <TIAACK>

Data Item : TIAACK Equipment acknowledgement code, 1 byte

0	= everything correct
1	= too many SVIDs (Max :100)
2	= no more traces allowed (Max :10)
3	= invalid period (DSPER error)
4	= invalid SVID
5	= invalid TRID
65	= System Error

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S2F25 Loopback Diagnostic Request(LDR) S,H<->E,reply

Description: A diagnostic message for checkout of protocol and communication circuits. The binary string sent is echoed back.

Structure: <ABS>

Data Item : ABS Any binary string

S2F26 Loopback Diagnostic Data(LDD) S,H<->E

Description: The echoed binary string of S2F25

Structure: <ABS>

Data Item : ABS Any binary string

S2F29 Equipment Constant Namelist Request(ECNR) S,H->E,reply

Description: Host can request the basic information of the enabled equipment constants.

Structure: L,n
1.<ECID1>
.
.
n.<ECIDn>

Exception: A zero length list (n=0) means send information for all ECIDs.

Data Item : ECID Equipment const ID

S2F30 Equipment Constant Namelist(ECN) M,H<-E

Description: Data response of S2F29

Structure: L,n (number of equipment constants)
1.L,6
1.<ECID1>
2.<ECNAME1>
3.<ECMIN1>
4.<ECMAX1>
5.<ECDEF1>
6.<UNITS1>
.
.
.
n.L,6
1.<ECIDn>
2.<ECNAMEn>
3.<ECMINn>
4.<ECMAXn>
5.<ECDEFn>
6.<UNITSn>

Exception If ECNAME,ECMIN,ECMAX,ECDEF of UNITS is zero length item, that ECID dose not exist.

Data Item : ECID Equipment const ID
ECNAME Equipment const Name
ECMIN Equipment const Minimum
ECMAX Equipment const Maximum
ECDEF Equipment const Default
UNITS Unit and Note

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S2F31 Date and Time Set Request(DTS) S,H->E,reply

Description: Useful to synchronize the equipment time with the host time base.

Structure: <TIME>

Data Item : TIME The length is different depending on equipment constant (ECV:TimeFormat)

TimeFormat	Length	Format
0	12byte	YYMMDDhhmmss
1	16byte	YYYYMMDDhhmmsscc
2	14byte	YYYYMMDDhhmmss

YYYY	= Year	000	～	9999
		0		
MM	= Month	01	～	12
DD	= Day	01	～	31
hh	= Hour	00	～	59
mm	= Minuites	00	～	59
ss	= Second	00	～	59
cc	= 1/100 sec	00	～	99

note: This is format and is not precision

S2F32 Date and Time Set Acknowledge(DTA) S,H<-E

Description: Acknowledge the receipt of time and date.

Structure: <TIACK>

Data Item : TIACK Time acknowledge code, 1byte
0 = OK
1 = Error, not done
65 = System Error

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S2F33 Define Report(DR) M,H->E,reply

Description: The purpose of this message is for the host to define a group of reports for the equipment. If S2F33 is Multi-block, it must be preceded by the S2F39/S2F40 Inquire/Grant transaction..

Structure:
```
L,2
  1.<DATAID>
  2.L,a
    1.L,2
      1.<RPTID1>
      2.L,b
        1.<VID1>
        .
        .
        b.<VIDb>
    .
    .
    a.L,2
      1.<RPTIDa>
      2.L,c
        1.<VID1>
        .
        .
        c.<VIDc>
```

Exception:
1. A list of zero length (a=0) following <DATAID> deletes all report definitions and associated links. See S2F35(Link Event/Report)
2. A list of zero length (b=0/c=0) following <RPTID> deletes report type RPTID. All CEID links to this RPTID are also deleted.

Data Item :

DATAID	Data ID
RPTID	Report ID
VID	Variable ID

S2F34 Define Report Acknowledge(DRA) S,H<-E

Description: Acknowledge or error. If an error condition is detected the entire message is rejected, i.e. partial changes are not allowed.

Structure: <DRACK>

Data Item : DRACK Define Report Acknowledge Code, 1 byte
- 0 = Accepted
- 1 = Denied. Insufficient space
- 2 = Denied. Invalid format
- 3 = Denied. At least one RPTID already defined
- 4 = Denied. At least one VID does not exist
- 65 = System Error

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S2F35 Link Event Report(LER) M,H->E,reply

Description: The purpose of this message is for the host to link n reports to an event(CEID). These linked event reports will default to 'disabled' upon linking. That is, the occurrence of an event would not cause the report to be sent until enabled. See S2F37 for enabling reports.

Structure:
```
L,2
  1.<DATAID>
  2.L,a
    1.L,2
      1.<CEID1>
      2.L,b
        1.<RPTID1>
      .
      .
        b.<RPTIDb>
    .
    .
    a.L,2
      1.<CEIDa>
      2.L,c
        1.<RPTID1>
        .
        .
        c.<RPTIDc>
```

Exception: A list of zero length (a=0) deletes all links of all event
A list of zero length (b=0) deletes all links of that event

Data Item :

	DATAID	Data ID
	CEID	Collection Event ID
	RPTID	Report ID
	VID	Variable ID

S2F36 Link Event Report Acknowledge(LERA) S,H<-E

Description: Acknowledge or error of S2F35. If an error condition is detected the entire message is rejected, i.e. partial changes are not allowed.

Structure: <LRACK>

Data Item :

	LRACK	Link Report Acknowledge Code,1byte
		0 = Accepted
		1 = Denied. Insufficient space
		2 = Denied. Invalid format
		3 = Denied. At least one CEID link already defined
		4 = Denied. At least one CEID does not exist
		5 = Denied. At least one RPTID does not exist
		65 = System Error

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S2F37 Enable/Disable Event Report(EDER) S,H->E,reply

Description: The purpose of this message is for the host to enable or disable reporting for a group of events (CEIDs)

Structure:
```
L,2
  1.<CEED>
  2.L,n
    1.<CEID1>
    .
    .
    n.<CEIDn>
```

Exception: A list of zero length (n=0) means all CEIDs

Data Item :
- CEED Collection event enable/disable code, 1 byte
  - FALSE = Disable
  - TRUE = Enable
- CEID Collection Event ID

S2F38 Enable/Disable Event Report Acknowledge(EERA) S,H<-E

Description: Acknowledge or error of S2F37. If an error condition is detected the entire message is rejected, i.e. partial changes are not allowed.

Structure: <ERACK>

Data Item :
- ERACK Enable/Disable Event Report Acknowledge Code, 1 byte
  - 0 = Accepted
  - 1 = Denied. At least one CEID does not exist
  - 65 = System Error

S2F39 Multi-block Inquire (DMBI) S,H->E,reply

Description: If a S2F33,S2F35 or S2F45 message is more than one block, this transaction must precede the message.

Structure:
```
L,2
  1.<DATAID>
  2.<DATALENGTH>
```

Data Item :
- DATAID Data ID
- DATALENGTH Total bytes to be sent

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S2F40 Multi-block Grant(MBG) S,H<-E

Description: Grant permission to send multi-block message

Structure: <GRANT>

Data Item : GRANT Grant code, 1 byte
0 = positive response, load OK
1 = busy, try again
2 = no space
3 = duplicate name
65 = System Error

S2F41 Host Command Send(HCS) S,H->E,reply

Description: The Host requests the Equipment perform the specified remote command with the associated parameters

Structure: L,2
1.<RCMD>
2.L,n (n is parameter count)
1.L,2
1.<CPNAME1>
2.<CPVAL1>
.
.
n.L,2
1.<CPNAMEn>
2.<CPVALn>

Data Item : RCMD Remote Command Name
CPNAME Command Parameter Name
CPVAL Command Parameter Value

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S2F42 Host Command Acknowledge(HCA) S,H<-E

Description: Acknowledge Host command or error. If command is not accepted due to one or more invalid parameters. (i.e. HCACK = 3), then a list of invalid parameters will be returned containing the parameter name and reason for being invalid.

Structure:
```
L,2
  1.<HCACK>
  2.L,n# of parameters
    1.L,2
      1.<CPNAME1>
    2.<CPACK1>
    .
    .
    n.L,2
      1.<CPNAMEn>
      2.<CPACKn>
```

Exception: If there are no invalid parameters then a list of zero length will be sent for item 2.

Data Item :
- HCACK Host Command Parameter Acknowledge Code, 1byte
  - 0 = Acknowledge, command has been performed
  - 1 = Command does not exist
  - 2 = Cannot perform now
  - 3 = At least one parameter is invalid
  - 4 = Acknowledge, command will be performed with completion signaled later by an event.
  - 5 - 63 = Reserved
- CPNAME Command Parameter Name
- CPACK Command Parameter Acknowledge Code, 1byte
  - 1 = Parameter Name(CPNAME) does not exist
  - 2 = Illegal Value specified for CPVAL
  - 3 = Illegal Format specified for CPVAL
  - >3 = Other equipment-specific error
  - 4 - 63 = reserved

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S2F43 Reset Spooling Streams and Functions(RSSF) S,H->E,reply

Description: If the spooling function is enabled (ECV: EnableSpooling=TRUE)、
Host change enable/disable for the each stream no. and function no.

When Host use this message, the only selected S/Fs are enabled.
(the not selected S/Fs are disabled.)

It is possible to Spool S/F
S2F17
S5F1
S6F1
S6F11
S7F3
S7F5
S10F1

Structure:
```
L,m
  1.L,2
    1.<STRID1>
    2.L,n
      1.<FCNID1>
    .
    .
      n.<FCNIDn>
  .
  .
  m.L,2
    1.<STRIDm>
    2.L,n
      1.<FCNID1>
    .
    .
      n.<FCNIDn>
```

Exception:
1. A zero length list (m = 0), turns off spooling for all streams and functions.
2. A zero length list (n = 0), turns on spooling for all functions for the associated stream.

Note:
1. Spooling for Stream 1 is not allowed..

Data Item :
STRID Stream No.
FCNID Function No.

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S2F44 Reset Spooling Acknowledge(RSA) M,H<-E

Description: Acknowledge or error

```
Structure:   L,2
               1.<RSPACK>                  (accept or reject)
               2.L,m                       (m = number of streams with errors)
                 1.L,3
                   1.<STRID1>              (Stream No)
                   2.<STRACK1>             (error in stream)
                   3.L,n                   (n = number of functions in error)
                     1.<FCNID1>            (Function No.)
                     .
                     .
                     n.<FCNIDn>
                 .
                 .
                 m.L,3
                   1.<STRIDm>
                   2.<STRACKm>             (error in stream)
                   3.L,n                   (n = number of functions in error)
                     1.<FCNID1>
                     .
                     .
                     n.<FCNIDn>
```

Exception:
1. If RSACK = 0, a zero length list (m = 0) is given, indicating no streams or functions in error.
2. A zero length list (n = 0) indicates no function in error for specified stream.

Data Item : RSPACK Reset Spooling Acknowledge, 1 byte
- 0 = Acknowledge, spooling setup accepted
- 1 = Spooling setup is rejected

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S2F45 Define Variable Limit Attributes(DVLA) M,H->E,reply

Description: Host can define the limit attributes for the specific VID
8 Limits can be defined for each specific VID

Structure:

```
L,2
  1.<DATAID>
    Data ID
  2.L,m                    (m = number of variables in this definition)
    1.L,2
      1.<VID1>
        Variable ID
      2.L,n                (n = number of limits for VID1)
        1.L,2
          1.<LIMITID1>
          2.L,p            (p=0 or 2)
            1.<UPPERDB1>
            2.<LOWERDB1>
        .
        .
        n.L,2
          1.<LIMITIDn>
          2.L,p            (p=0 or 2)
            1.<UPPERDB1>
            2.<LOWERDB1>
    .
    .
    m.L,2
      1.<VIDm>
      2.L,n                 (n = number of limits for VIDm)
        1.L,2
          1.<LIMITID1>
          2.L,p            (p=0 or 2)
            1.<UPPERDB1>
            2.<LOWERDB1>
        .
        .
        n.L,2
          1.<LIMITIDn>
          2.L,p            (p=0 or 2)
            1.<UPPERDB1>
            2.<LOWERDB1>
```

Exception:

1. A zero length list (m = 0) sets all limit values for all monitored VIDs to "undefined".
2. A zero length list (n = 0) sets all limits values for that VID to "undefined"
3. A zero length list (p = 0) sets that limit to "undefined".

Data Item :

LIMITID	Limit ID each specific VID、1byte
UPPERDB	Upper Limit attribute
LOWERDB	Lower Limit attribute

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S2F46 Variable Limit Attribute Acknowledge(VLAA) M,H<-E

Description: Acknowledge definition of variable limit attributes or report error. If DVLA is not accepted due to one or more invalid parameters (e.g.LIMITACK=3), then a list of invalid parameters is returned containing the variable limit attribute and reason for rejection. If an error condition is detected, the entire message is rejected, i.e. partial changes are not allowed.

Structure:
```
L,2
  1.<VLAACK>
  2.L,m                  (m = number of invalid parameters)
    1.L,3
      1.<VID1>           (VID with error)
      2.<LVACK1>         (reason)
      3.L,n              (n=0 or 2)
        1.<LIMITID1>     (1st limit in error for VIDp)
        2.<LIMITACK1>    (reason)
    .
    .
    m.L,3
      1.<VIDm>           (VID with error)
      2.<LVACKm>         (reason)
      3.L,n              (n=0 or 2)
        1.<LIMITIDm>     (1st limit in error for VIDp)
        2.<LIMITACKm>    (reason)
```

Exception:
1. A zero length list (m=0) indicates no invalid variable limit attributes
2. A zero length list (n=0) indicates no invalid limit values for that VID.

Data Item :

VLAACK Variable Limit Attribute Acknowledge Code, 1byte
- 0 = Acknowledge, command will be performed
- 1 = Limit attribute definition error
- 2 = Cannot perform now
- 65 = System Error

LVACK Variable limit definition acknowledges code, 1byte.
- 1 = Variable does not exist.
- 2 = Variable has no limits capability.
- 3 = Variable repeated in message.
- 4 = Limit value error as described in LIMITACK.

LIMITID Limit ID each specific VID、1byte

LIMITACK Acknowledgment code for variable limit attribute set, 1byte
- 1 = LIMITID does not exist.
- 2 = UPPERDB > LIMITMAX.
- 3 = LOWERDB < LIMITMIN.
- 4 = UPPERDB < LOWERDB.
- 5 = Illegal format specified for UPPERDB or LOWERDB.
- 6 = ASCII value limit definition for this variable.

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S2F47 Variable Limit Attribute Request(VLAR)	S,H->E,reply

Description: This message allows the host to query the equipment for current variable limit attribute definitions.

Structure: L,m
1.<VID1>
.
.
m.<VIDm>

Exception: A zero length list (m=0), requests a list of all VID values that can have variable limit attributes.

Data Item : VID Variable ID

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S2F48 Variable Limit Attributes Send(VLAS) M,H<-E

Description: Equipment sends values of requested variable limit attribute definitions in the order requested.

Structure:

```
L,m
  1.L,2
    1.<VID1>                    (VID1)
    2.L,p                       (P=0 or 4)
      1.<UNITS1>
      2.<LIMITMIN1>
      3.<LIMITMAX1>
      4.L,n
        1.L,3
          1.<LIMITID1>
          2.<UPPERDB1>
          3.<LOWERDB1>
        .
        .
        n.L,3
          1.<LIMITIDn>
          2.<UPPERDBn>
          3.<LOWERDBn>
  .
  .
    m.L,2
      1.<VIDm>                   (VIDm)
      2.L,p                      (p=0 or 4)
        1.<UNITSm>
        2.<LIMITMINm>
        3.<LIMITMAXm>
        4.L,n
          1.L,3
            1.<LIMITID1>
            2.<UPPERDB1>
            3.<LOWERDB1>
          .
          .
          n.L,3
            1.<LIMITIDn>
            2.<UPPERDBn>
            3.<LOWERDBn>
```

Exception:

1. A zero length list (p=0) indicates that limits are not supported for the VID.
2. A zero length list (n=0) means no limits are currently defined for the specified variable.

Data Item :

UNITS	Units and Note
LIMITMIN	The minimum allowed value for the limit values of a specific variable.
LIMITMAX	The maximum allowed value for the limit values of a specific variable
LIMITID	The identifier of a specific limit in the set of limits for a variable
UPPERDB	The variable limit attribute which defines the upper boundary of the deadband of a limit.
LOWERDB	The variable limit attribute which defines the lower boundary of the deadband of a limit.

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6-3. Stream5 Exception Handling

The called alarm means alarm or error of the equipment.

S5F0 Abort Transaction(S5F0) S,H<->E

Description: Same from as S1F0

S5F1 Alarm Report Send(ARS) S,H<-E,reply

Description: This message reports a change in or presence of an alarm condition. One message will be issued when the alarm is set and one message will be issued when the alarm is cleared. Irrecoverable errors and attention flags may not have a corresponding clear message.

Structure: L,3
1.<ALCD>
2.<ALID>
3.<ALTX>

Data Item : ALCD Alarm code byte
bit8 = 1 means alarm set
bit8 = 0 means alarm cleared
bits7 - 1 is alarm category code = not used
ALID Alarm identification
ALTX Alarm text

S5F2 Alarm Report Acknowledge(ARA) S,H->E

Description: Acknowledge or error of S5F1

Structure: <ACKC5>

Data Item : ACKC5 Acknowledge Code, 1byte
0 = accepted
64 = Invalid ALID
65 = System Error

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S5F3 Enable/Disable Alarm Send(EAS) S,H->E,reply

Description: Host can change the enable /disable of specific alarm.
The equipment send S5F1 of the alarm which is enabled.

Structure: L,2
1.<ALED>
2.<ALID>

Exception: A zero length item for ALID means all alarms

Data Item : ALED Alarm enable/disable code, 1byte
bit8 = 1 means enable alarm
bit8 = 0 means disable alarm
ALID Alarm identification

S5F4 Enable/Disable Alarm Acknowledge(EAA) S,H<-E

Description: Acknowledge or error of S5F3

Structure: <ACKC5>
Data Item : ACKC5 Acknowledge Code, 1byte
0 = accepted
64 = Invalid ALID
65 = System Error

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S5F5 List Alarm Request(LAR) S,H->E,reply

Description: This message requests the equipment to send binary and analog alarm information to the host.

Structure: <ALID1,...,ALIDn>

Exception: A zero length item means send all possible alarms regardless of the state of ALED

S5F6 List Alarm Data(LAD) M,H<-E

Description: The equipment send the informations of the alarm selected by S5F5

Structure:
```
L,m
  1.L,3
    1.<ALCD1>
    2.<ALID1>
    3.<ALTX1>
  .
  .
  m.L,3
    1.<ALCDm>
    2.<ALIDm>
    3.<ALTXm>
```

Exception: If m = 0, no response can be made. A zero length item returned for ALCDi or ALTXi means that value does not exist.

Data Item :
```
ALCD   Alarm code byte
                bit8 = 1 means alarm set
                bit8 = 0 means alarm cleared
                bits7 - 1 is alarm category code = not used
ALID   Alarm identification
ALTX   Alarm text
```

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6-4. Stream6 Data Collection

S6F0 Abort Transaction(S6F0) S,H<->E

Description: Same from as S1F0

S6F1 Trace Data Send(TDS) S,H<-E,reply

Description: This function sends samples to the host according to the trace setup done by S2F23. Trace is a time-driven form of equipment status.

Structure:
```
L,4
  1.<TRID>
  2.<SMPLN>
  3.<STIME>
  4.L,n
    1.<SV1>
    2.<SV2>
    .
    .
    n.<SVn>
```

Exception: A zero length STIME means no value is given and that the time is to be derived from SMPLN along with knowledge of the request.

Data Item :
- TRID Trace request ID
- SMPLN The last sample number in this message (acutual sample count)
- STIME The last sample time in this message The length is different depending on equipment constant (ECV:TimeFormat)

TimeFormat	Length	Format
0	12byte	YYMMDDhhmmss
1	16byte	YYYYMMDDhhmmsscc
2	14byte	YYYYMMDDhhmmss

YYYY	= Year	000	～	9999
MM	= Month	01	～	12
DD	= Day	01	～	31
hh	= Hour	00	～	59
mm	= Minuites	00	～	59
ss	= Second	00	～	59
cc	= 1/100 sec	00	～	99

note: This is format and is not precision

S6F2 Trace Data Acknowledge(TDA) S,H->E

Description: Acknowledge or error of S6F1

Structure: <ACKC6>

Data Item : ACKC6 Acknowledge Code, 1byte
- 0 = accepted and done
- 65 = System Error

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S6F5 Multi-block Data Send Inquire(MBI) S,H<-E,

Description: If the discrete data report S6F11 can involve more than one block, this transaction must precede the transmission.

Structure:
```
L,2
  1.<DATAID>
  2.<DATALENGTH>
```

Data Item :

DATAID	Data ID
DATALENGTH	Total bytes to be sent

S6F6 Multi-block Grant(MBG) S,H->E

Description: Acknowledge or error of S6F5

Structure: <GRANT6>

Data Item : GRANT6 Permission to send, 1byte
- 0 = Permission granted
- 1 = Busy, try again
- 2 = Not interested
- 65 = System Error

S6F11 Event Report Send(ERS) M,H<-E,reply

Description: The purpose of this message is for the equipment to send a defined, linked and enabled group of reports to the host upon the occurrence of an event(CEID). If S6F11 is Multi-block(over 254byte), it must be preceded by the S6F5/S6F6 Inquire/Grant transaction.

Structure:
```
L,3
  1.<DATAID>
  2.<CEID>
  3.L,a
    1.L,2                                (レポート 1)
      1.<RPTID1>
      2.L,b
        1.<V1>
        .
        .
        b.<Vb>
    .
    .
    1.L,2                                (レポート a)
      1.<RPTIDa>
      2.L,c                          (レポート内の V の数)
        1.<V1>
        .
        .
        c.<Vc>
```

Exception: If there are no reports linked to the event a 'null' report is assumed. A zero length list for # of reports means there are no reports linked to the given CEID

Data Item :

DATAID	Data ID
CEID	Collection Event ID
RPTID	Report ID
V	Variable Data(SV、ECV or DV)

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S6F12 Event Report Acknowlege(ERA) S,H->E

Description: Acknowledge or error of S6F11

Structure: <ACKC6>

Data Item : ACKC6 Acknowledge Code, 1byte
0 = accepted
65 = System Error

S6F15 Event Report Request(ERR) S,H->E,reply

Description: Host can request the reports linked to specific event by Sending CEID

Structure: <CEID>

Data Item : CEID Collection Event ID

S6F16 Event Report Data(ERD) M,H<-E

Description: Equipment sends reports linked to given CEID to host.

Structure: Same from as S6F11

Exception: Same from as S6F11

S6F19 Individual Report Request(IRR) S,H->E,reply

Description: Host can request a specific report by Sending RPTID

Structure: <RPTID>

Data Item : RPTID Report ID

S6F20 Individual Report Data(IRD) M,H<-E

Description: Equipment sends variable data defined for the given RPTID to the host.

Structure: L,n
1.<V1>
.
.
n.<Vn>

Exception: A zero length list (n=0) means RPTID is not defined.

Data Item : V Variable Data(SV、ECV or DV)

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S6F23 Request Spooled Data(RSD) S,H->E,reply

Description: The purpose of this message is for the host to request transmission or deletion of the messages currently spooled by the equipment.

Structure: <RSDC>

Data Item : RSDC Request Spool Data Code
0 = Transmit Spooled Messages
1 = Purge Spooled Messages

S6F24 Request Spooled Data Acknowledgement Send(RSDAS) S,H<-E

Description: Acknowledge or error of S6F23

Structure: <RSDA>

Data Item : RSDA Request Spool Data Acknowledge
0 = OK
1 = Denied, busy try later
2 = Denied, spooled data does not exist
65 = System Error

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6-5. Stream7 Process Program Management

S7F0 Abort Transaction(S7F0) S,H<->E

Description: Same from as S1F0

S7F1 Process Program Load Inquire (PPI) S,H<->E,reply

Description: This message is used to initiate the transfer of a process program(S7F3/S7F4 or S7F23/S7F24).

Structure: L,2
1.<PPID>
2.<LENGTH>

Data Item : PPID Process program ID
LENGTH Length of the process program in bytes

S7F2 Process Program Load Grant (PPG) S,H<->E

Description: This message gives permission for the process program to be loaded

Structure: <PPGNT>

Data Item : PPGNT Process program grant status, 1byte
0 = OK
1 = OK (already have)
2 = no space
3 = invalid PPID
4 = busy, try later
5 = will not accept

S7F3 Process Program Send(PPS) M,H<->E,reply

Description: The program is sent.
When the S7F3 is multi-block、S7F1/F2 transaction should be sent before this.
(Optional : S7F1/F2 can be omitted)

Structure: L,2
1.<PPID>
2.<PPBODY>

Data Item : PPID Process program ID
PPBODY Process program body

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S7F4 Process Program Acknowledge(PPA) S,H<->E

Description: Acknowledge or error of S7F3

Structure: <ACKC7>

Data Item : ACKC7 Acknowledge Code, 1byte
0 = Accepted
1 = Permission not granted
2 = Length error
3 = Matrix overflow
4 = PPID not found
5 = Mode unsupported
6 - 63 = reserved

S7F5 Process Program Request(PPR) S,H<->E,reply

Description: This message is used to request the transfer of a process program.

Structure: <PPID>

Data Item : PPID Process program ID

S7F6 Process Program Data(PPD) M,H<->E

Description: This message is used to transfer a process program.

Structure: L,2
1.<PPID>
2.<PPBODY>

Exception: A zero length list means request denied

Note: The equipment to host transfer of the process program, denoted by the R bit in the header (R = 1), provides the mechanism for the host computer to receive process programs created on the equipment. This allows use of the equipment without having process program generation capabilities on the host.

Data Item : PPID Process program ID
PPBODY Process program body

S7F17 Delete Process Program Send(DPS) S,H->E,reply

Description: This message is used by the host to request the equipment to delete process programs from equipment storage.

Structure: L,n (Number of process programs to be deleted)
1.<PPID1>
.
.
n.<PPIDn>

Exception: If n = 0, then delete all.

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S7F18 Delete Process Program Acknowledge(DPA) S,H<-E

Description: Acknowledge or error of S7F18

Structure: <ACKC7>

Data Item : ACKC7 Acknowledge Code, 1byte
0 = Accepted
1 = Permission not granted
2 = Length error
3 = Matrix overflow
4 = PPID not found
5 = Mode unsupported
6 - 63 = reserved

S7F19 Current EPPD Request(RER) S,H->E,reply

Description: This message is used to request the transmission of the current equipment process program directory(EPPD). This is a list of all the PPIDs of the process programs stored in the equipment.

Structure: Header only

S7F20 Current EPPD Data(RED) M,H<-E

Description: This message is used to transmit the current EPPD.

Structure: L,n (number of all process programs in the equipment)
1.<PPID1>
.
.
n.<PPIDn>

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S7F23 Formatted Process Program Send(FPS) M,H<->E,reply

Description: The formatted process program is sent.
When the S7F23 is multi-block、S7F1/F2 transaction should be sent before this.
(Optional : S7F1/F2 can be omitted)

Structure:
```
L,4
  1.<PPID>
  2.<MDLN>
  3.<SOFTREV>
  4.L,c                          (recipe item count)
   1.L,2
    1.<CCODE>
    2.L,7                        (7 fix)
      1.<PPARM1>
      2.<PPARM2>
      3.<PPARM3>
      4.<PPARM4>
      5.<PPARM5>
      6.<PPARM6>
      7.<PPARM7>
   .
   .
   c.L,2
    1.<CCODE>
    2.L,7                        (7 fix)
      1.<PPARM1>
      2.<PPARM2>
      3.<PPARM3>
      4.<PPARM4>
      5.<PPARM5>
      6.<PPARM6>
      7.<PPARM7>
```

Data Item :

PPID	Process program ID
MDLN	Equipment Model Type
SOFTREV	Software revision code
CCODE	Specific ID for each recipe item
PPARM1	Use or No use of the recipe item(ASC:“TRUE” or”FALSE”)
PPARM2	Recipe item Name(ASC)
PPARM3	Recipe item Unit(ASC)
PPARM4	Recipe item Format(ASC)
PPARM5	Recipe item value(F8)
PPARM6	Recipe item minimum(F8)
PPARM7	Recipe item maximum (F8)

S7F24 Formatted Process Program Acknowledge(FPA) M,H<->E

Description: Acknowledge or error of S7F23

Structure: <ACKC7>

Data Item : ACKC7 Acknowledge Code, 1byte
- 0 = Accepted
- 1 = Permission not granted
- 2 = Length error
- 3 = Matrix overflow
- 4 = PPID not found
- 5 = Mode unsupported
- 6 - 63 = reserved

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S7F25 Formatted Process Program Request(FPR) M,H<->E,reply

Description: This message is used to request the transfer of a formatted process program.

Structure: <PPID>

Data Item : PPID Process Program ID

S7F26 Formatted Process Program Data(FPD) M,H<->E

Description: This message is used to transfer a formatted process program.

Structure:
```
L,4
  1.<PPID>
  2.<MDLN>
  3.<SOFTREV>
  4.L,c                          (recipe item count)
   1.L,2
    1.<CCODE>
    2.L,7                        (7 fix)
      1.<PPARM1>
      2.<PPARM2>
      3.<PPARM3>
      4.<PPARM4>
      5.<PPARM5>
      6.<PPARM6>
      7.<PPARM7>
   .
   .
   c.L,2
    1.<CCODE>
    2.L,7                        (7 fix)
      1.<PPARM1>
      2.<PPARM2>
      3.<PPARM3>
      4.<PPARM4>
      5.<PPARM5>
      6.<PPARM6>
      7.<PPARM7>
```

Exception : A zero length list (c=0) means request denied

Data Item :

PPID	Process program ID
MDLN	Equipment Model Type
SOFTREV	Software revision code
CCODE	Specific ID for each recipe item
PPARM1	Use or No use of the recipe item(ASC:“TRUE” or”FALSE”)
PPARM2	Recipe item Name(ASC)
PPARM3	Recipe item Unit(ASC)
PPARM4	Recipe item Format(ASC)
PPARM5	Recipe item value(F8)
PPARM6	Recipe item minimum(F8)
PPARM7	Recipe item maximum (F8)

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S7F27 Process Program Verification Send(PVS) M,H←E,reply

Description: The equipment send the result of checking the received the formatted process program.

Structure: **■When parameter abnormal (Process Program Send Fail)**

```
L,2
  1.<PPID>
  2.L,n                (number of error)
   1.L,3
      1.<ACKC7A>
      2.<SEQNUM>
      3.<ERRW7>
   2.L,3
   .
   .
   n.L,3
```

■When parameter normal (Process Program Send Success)

```
L,2
  1.<PPID>
  2.L,0
```

Data Item :

PPID	Process program ID
ACKC7A	Acknowledge Code, 1byte 0 = Accepted 1 = MDLN unmatch 2 = SOFTREV unmatch 3 = CCODE error 4 = PPARM error 5 = Other(report by ERRW7)
SEQNUM	Uniq number in the list This number starts from 1
ERRW7	The reason text of errors

S7F28 Process Program Verification Acknowledge(PVA) M,H→E

Description: Acknowledge of S7F27

Structure: Header only

S7F29 Process Program Verification Inquire(PVI) M,H←E,reply

Description: The equipment request to send the multi-block PVS

Structure: <LENGTH>

Data Item : LENGTH The length of S7F27

S7F30 Process Program Verification Grant(PVG) M,H→E

Description: Acknowledge or error of S7F29

Structure: <PPGNT>

Data Item :

PPGNT	Process program grant status, 1byte 0 = OK 1 = OK (already have) 2 = no space 3 = invalid PPID 4 = busy, try later 5 = will not accept

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6-6. Stream9 System Errors

S9F0 Abort Transaction(S9F0) **S,H<->E**

Description: Same from as S1F0

S9F1 Unrecognized Device ID(UDN) **S,H<-E**

Description: The device ID in the message block header did not correspond to any known device ID in the node detecting the error.

Structure: <MHEAD>

Data Item : MHEAD SECS message block header associated with message block in error

S9F3 Unrecognized Stream Type(USN) **S,H<-E**

Description: The equipment does not recognize the stream type in the message block header.

Structure: <MHEAD>

Data Item : MHEAD SECS message block header associated with message block in error

S9F5 Unrecognized Function Type(UFN) **S,H<-E**

Description: This message indicates that the function in the message ID is not recognized by the receiver.

Structure: <MHEAD>

Data Item : MHEAD SECS message block header associated with message block in error

S9F7 Illegal Data(IDN) **S,H<-E**

Description: This message indicates that the stream and function were recognized but the associated data format could not be interpreted.

Structure: <MHEAD>

Data Item : MHEAD SECS message block header associated with message block in error

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S9F9 Transaction Timer Timeout(TTN) S,H<-E

Description: This message indicates that a transaction(receive) timer has timed out and that he corresponding transaction has been aborted. It is up to respond to this error in an appropriate to keep the system operational.

Structure: <SHEAD>

Data Item : SHEAD Stored header related to the transaction timer

S9F11 Data Too Long(DLN) S,H<-E

Description: This message to the host indicates that the equipment has been sent more data that it can handle.

Structure: <MHEAD>

Data Item : MHEAD SECS message block header associated with message block in error

S9F13 Conversation Timeout(CTN) S,H<-E

Description: Though data should have been received, Host was not able to receive it before appropriate time.
The resource is cleared.

Structure: L,2
1.<MEXP>

2.<EDID>

Data Item : MEXP The message which the equipment should receive SxxFyy
EDID The data ID which the equipment should receive
In the case of S7F03 or S7F23, this is PPID

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6-7. Stream10 Terminal Services

S10F0 Abort Transaction(S10F0) S,H<->E

Description: Same from as S1F0

S10F1 Terminal Request(TRN) S,H<-E,reply

Description: A terminal text message to the host.

Structure: L,2
1.<TID>
2.<TEXT>

Data Item : TID Terminal number, 1byte
0 = single or main terminal
0 <> addtional terminals at the same equipment
TEXT A single line of characters

S10F2 Terminal Request Acknowledge(TRA) S,H->E

Description: Acknowledge or error of S10F1

Structure: <ACKC10>

Data Item : ACKC10 Acknowledge Code, 1byte
0 = Accepted for display
1 = Message will not be displayed
2 = Terminal not available
65 = System Error

S10F3 Terminal Display, Single(VTN) S,H->E,reply

Description: Data to be displayed.

Structure: L,2
1.<TID>
2.<TEXT>

Data Item : TID Terminal number, 1byte
0 = single or main terminal
0 <> addtional terminals at the same equipment
TEXT A single line of characters

S10F4 Terminal Display, Single Acknowledge(VTA) S,H<-E

Description: Acknowledge or error of S10F3

Structure: <ACKC10>

Data Item : ACKC10 Acknowledge Code, 1byte
0 = Accepted for display
1 = Message will not be displayed
2 = Terminal not available
65 = System Error

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S10F5 Terminal Display, Multi-block(VTN) S,H->E,reply

Description: Data to be displayed.

Structure:
```
L,2
  1.<TID>
  2.L,n
    1.<TEXT1>
    .
    .
    n.<TEXTn>
```

Data Item :
- TID — Terminal number, 1byte
  - 0 = single or main terminal
  - 0 <> addtional terminals at the same equipment
- TEXT — A single line of characters

S10F6 Terminal Display, Multi-block Acknowledge(VTA) S,H<-E

Description: Acknowledge or error of S10F5

Structure: <ACKC10>

Data Item :
- ACKC10 — Acknowledge Code, 1byte
  - 0 = Accepted for display
  - 1 = Message will not be displayed
  - 2 = Terminal not available
  - 65 = System Error

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7 . Data Item Dictionary

7-1. Item Dictionary

This section defines the data items used in the standard SECS-II messages picked up for TOWA HOST COMMUNICATION SYSTEM.

The description of 'Name', 'Format', 'Description', 'Where' are specified with SEMI E5-0709E.

Especially, this section defines about 'Format' and data item length.

ABS **Format:10**

Any binary string

Where Used: S2F25,S2F26

ACKC5 **Format:10**

Acknowledge Code, 1byte
0 = Accepted
1 = invalid ALID
65 = System error

Where Used: S5F2,S5F4

ACKC6 **Format:10**

Acknowledge Code, 1byte
0 = Accepted and done
65 = System error

Where Used: S6F2,S6F12

ACKC7 **Format:10**

Acknowledge Code, 1byte
0 = Accepted
1 = Permission not granted
2 = Length error
3 = Matrix overflow
4 = PPID not found
5 = Mode unsupported
6 = Not REMOTE mode
6 - 63 = reserved

Where Used: S7F4,S7F18,S7F24

ACKC7A **Format:51**

Acknowledge Code, 1byte
Acknowledge Code, 1byte
0 = Accepted
1 = MDLN unmatch
2 = SOFTREV unmatch
3 = CCODE error
4 = PPARM error
5 = Other(report by ERRW7)

Where Used: S7F27

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ACKC10(確認コード) **Format:10**

Acknowledge Code, 1byte
0 = Accepted for display
1 = Message will not be displayed
2 = Terminal not available
65 = System error

Where Used: S10F2,S10F4,S10F6

ALCD **Format:10**

Alarm code byte
bit8 = 1 means alarm set
bit8 = 0 means alarm cleared
bits7 - 1 is alarm category code = not used

Where Used: S5F1,S5F6

ALED **Format:10**

Alarm enable/disable code, 1byte
bit8 = 1 means enable alarm
bit8 = 0 means disable alarm

Where Used: S5F3

ALID **Format:54**

Alarm identification

Where Used: S5F1,S5F3,S5F5,S5F6

ALTX **Format:20**

Alarm text

Where Used: S5F1,S5F6

CCODE **Format:11**

Specific ID for each recipe item

Where Used: S7F23,S7F26

CEED **Format:11**

Collection event enable/disable code, 1 byte
FALSE = Disable
TRUE = Enable

Where Used: S2F37

CEID **Format:52,54**

Collected event ID

Where Used: S2F35,S2F37,S6F11,S6F15,S6F16

COMMACK **Format:10**

Establish Communications Acknowledge Code, 1byte
0 = Accepted
1 = Denied, Try Again
65 = System error

Where Used: S1F14

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CPACK **Format:10**

Command Parameter Acknowledge Code, 1byte
1 = Parameter Name(CPNAME) does not exist
2 = Illegal Value specified for CPVAL
3 = Illegal Format specified for CPVAL
>3 = Other equipment-specific error
4 - 63 = reserved

Where Used: S2F42

CPNAME **Format:20**

COMMAND PARAMETER NAME

Where Used: S2F41,S2F42

CPVAL **Format:20, 5()**

Command Parameter Value

Where Used: S2F41

DATAID **Format:52,54**

Data ID

Where Used: S2F33,S2F35,S2F39,S2F45,S6F5,S6F11,S6F16

DATALENGTH **Format:54**

Total bytes to be sent

Where Used: S2F39,S6F5

DRACK **Format:10**

Define Report Acknowledge Code, 1 byte
0 = Accepted
1 = Denied. Insufficient space
2 = Denied. Invalid format
3 = Denied. At least one RPTID already defined
4 = Denied. At least one VID does not exist

Where Used: S2F34

DSPER **Format:20**

Data sample period
Minimum is 1 second
hhmmss, 6byte

Where Used: S2F23

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EAC **Format:10**

Equipment acknowledge code, 1byte
0 = Acknowledge
1 = Denied. At least one constant does not exist
2 = Denied. Busy
3 = Denied. At least one constant out of range
65 = System error

Where Used: S2F16

ECDEF **Format:10,11,20,21,3(),4(),5()**

Equipment constant default value

Where Used: S2F30

ECID **Format: 52,54**

Equipment Constant ID

Where Used: S2F13,S2F15,S2F29,S2F30

ECMAX **Format:10,11,20,21,3(),4(),5()**

Equipment constant maximum value

Where Used: S2F30

ECMIN **Format:10,11,20,21,3(),4(),5()**

Equipment constant minimum value

Where Used: S2F30

ECNAME **Format:20**

Equipment const Name

Where Used: S2F30

ECV **Format:10,11,20,21,3(),4(),5()**

Equipment const Value

Where Used: S2F14,S2F15

EDID **Format:10,20,3(),5()**

The data ID which the equipment should receive
In the case of S7F03 or S7F23, this is PPID

Where Used: S9F13

ERACK **Format:10**

Enable/Disable Event Report Acknowledge Code, 1 byte
0 = Accepted
1 = Denied. At least one CEID does not exist

Where Used: S2F38

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ERRW7 **Format:20**

The reason text of errors

Where Used: S7F27

FCNID **Format:51**

Function Identification

Where Used: S2F43,S2F44

GRANT **Format:10**

Grant code, 1 byte
0 = positive response, load OK
1 = busy, try again
2 = no space
3 = duplicate name
65 = System error

Where Used: S2F40

GRANT6 **Format:10**

Grant code, 1 byte
0 = Permission granted
1 = Busy, try again
2 = Not interested
65 = System error

Where Used: S6F6

HCACK **Format:10**

Host Command Parameter Acknowledge Code, 1byte
0 = Acknowledge, command has been performed
1 = Command does not exist
2 = Cannot perform now
3 = At least one parameter is invalid
4 = Acknowledge, command will be performed with completion signaled later by an event.
5 - 63 = Reserved

Where Used: S2F42

LENGTH **Format:54**

Length of the process program in bytes

Where Used: S7F1

LIMITACK **Format:10**

Acknowledgment code for variable limit attribute set, 1byte
1 = LIMITID does not exist.
2 = UPPERDB > LIMITMAX.
3 = LOWERDB < LIMITMIN.
4 = UPPERDB < LOWERDB.
5 = Illegal format specified for UPPERDB or LOWERDB.
6 = ASCII value limit definition for this variable.

Where Used: S2F46

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LIMITID **Format:10**

The identifier of a specific limit in the set of limits for a variable

Where Used: S2F45,S2F46,S2F48

LIMITMAX **Format:11,20,3(),4(),5()**

The maximum allowed value for the limit values of a specific variable. The equipment manufacturer should specify this value, which would typically coincide with the maximum value of the variable being monitored. The format must match that of the referenced variable.

Where Used: S2F48

LIMITMIN **Format:11,20,3(),4(),5()**

The minimum allowed value for the limit values of a specific variable. The equipment manufacturer should specify this value, which would typically coincide with the minimum value of the variable being monitored. The format must match that of the referenced variable.

Where Used: S2F48

LOWERDB **Format:11,20,3(),4(),5()**

The variable limit attributes which defines the lower boundary of the deadband of a limit. The value applies to a single limit (LIMITID) for a specified VID. Thus, UPPERDB and LOWERDB as a pair define a limit.

Where Used: S2F45,S2F48

LRACK **Format:10**

Link Report Acknowledge Code,1byte
0 = Accepted
1 = Denied. Insufficient space
2 = Denied. Invalid format
3 = Denied. At least one CEID link already defined
4 = Denied. At least one CEID does not exist
5 = Denied. At least one RPTID does not exist

Where Used: S2F36

LVACK **Format:10**

Variable limit definition acknowledges code, 1byte. Defines the error with the limit attributes for the referenced VID.
1 = Variable does not exist.
2 = Variable has no limits capability.
3 = Variable repeated in message.
4 = Limit value error as described in LIMITACK.

Where Used: S2F46

MDLN **Format:20**

Equipment Model Type
Same data as returned by S1F2.

Where Used: S1F2,S1F13,S1F14

MEXP **Format:20**

The message which you should receive SxxFyy.

Where Used: S9F13

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MHEAD **Format:10**

SECS message block header associated with message block in error

Where Used: S9F1,S9F3,S9F5,S9F7,S9F11

OFLACK **Format:10**

Acknowledge code for OFF-LINE request.
0 = OFF-LINE Acknowledge.
65 = System Error

Where Used: S1F16

ONLACK **Format:10**

Acknowledge code for ON-LINE request.
0 = ON-LINE Accepted
1 = Not grant
2 = Already ON-LINE
65 = System Error

Where Used: S1F18

PPARM **Format:11,20,4(),5()**

Process Parameter

Where Used: S7F23,S7F26

PPBODY **Format:10**

Process program body

Where Used: S7F3,S7F6

PPGNT **Format:10**

Process program grant status, 1byte
0 = OK
1 = OK (already have)
2 = no space
3 = invalid PPID
4 = busy, try later
5 = will not accept
6 - 63 = reserved

Where Used: S7F2

PPID **Format:20**

Process program ID

Where Used: S7F1,S7F3,S7F5,S7F6,S7F17,S7F20,S7F23,S7F24,S7F25,S7F26,S7F27,
S9F13

RCMD **Format:20**

REMOTE COMMAND Name

Where Used: S2F41

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REPGSZ **Format:54**

Report count every S6F1
When the equipment get the data this setting count,
S6F1 is sent.

Where Used: S2F23

RPTID **Format:52,54**

Report ID

Where Used: S2F33,S2F35,S6F11,S6F16,S6F19

RSPACK **Format:10**

Reset Spooling Acknowledge, 1 byte
0 = Acknowledge, spooling setup accepted
1 = Spooling setup rejected.

Where Used: S2F44

RSDA **Format:10**

Request Spool Data Acknowledge, 1 byte
0 = OK
1 = Denied, busy try later
2 = Denied, spooled data does not exist
65 = System Error

Where Used: S6F24

RSDC **Format:51**

Request Spool Data Code
0 = Transmit Spooled Messages
1 = Purge Spooled Messages

Where Used: S6F23

SEQNUM **Format:52**

Uniq number in the list
This number starts from 1

Where Used: S7F27

SHEAD **Format:10**

Stored header related to the transaction timer

Where Used: S9F9

SMPLN **Format:54**

The last sample number in this message
(acutual sample count)

Where Used: S6F1

SOFTREV **Format:20**

Software revision code

Where Used: S1F2,S1F13,S1F14

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STIME **Format:20**

The last sample time in this message
The length is different depending on equipment constant (ECV:TimeFormat)

TimeFormat	Length	Format
0	12byte	YYMMDDhhmmss
1	16byte	YYYYMMDDhhmmsscc
2	14byte	YYYYMMDDhhmmss

YYYY	= Year	0000	to	9999
MM	= Month	01	to	12
DD	= Day	01	to	31
hh	= Hour	00	to	59
mm	= Minuites	00	to	59
ss	= Second	00	to	59
cc	= 1/100 sec	00	to	99

Note: This is format and is not precision

Where Used: S6F1

STRACK **Format:10**

Spool Stream Acknowledge
1 = Spooling not allowed for stream(i.e., Stream 1).
2 = Stream unknown.
3 = Unknown function specified for this stream.
4 = Secondary function specified for this stream.

Note: Error codes 1 - 4 shown in order of precedence for reporting. Code 0 not used.

Where Used: S2F44

STRID **Format:51**

Stream Identification

Where Used: S2F43, S2F44

SV **Format:0,10,11,20,21,3(),4(),5()**

Status variable value

Where Used: S1F4, S6F1

SVID **Format:52,54**

Status variable ID

Where Used: S1F3, S1F11, S1F12, S2F23

SVNAME **Format:20**

Status Variable Name

Where Used: S1F12

TEXT **Format:20**

A single line of characters

Where Used: S10F1, S10F3

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TIAACK **Format:10**

Equipment acknowledgement code, 1 byte
0 = everything correct
1 = too many SVIDs
2 = no more traces allowed
3 = invalid period
65 = System Error

Where Used: S2F24

TIACK **Format:10**

Time Acknowledge Code, 1 byte
0 = OK
1 = Error, not done
65 = System Error

Where Used: S2F24

TID **Format:10**

Terminal number, 1byte
0 = single or main terminal
>0 = additional terminals at the same equipment

Where Used: S10F1, S10F3

TIME **Format:20**

The length is different depending on equipment constant (ECV:TimeFormat)

TimeFormat	Length	Format
0	12byte	YYMMDDhhmmss
1	16byte	YYYYMMDDhhmmsscc
2	14byte	YYYYMMDDhhmmss

YYYY = Year 0000 to 9999
MM = Month 01 to 12
DD = Day 01 to 31
hh = Hour 00 to 59
mm = Minuites 00 to 59
ss = Second 00 to 59
cc = 1/100 sec 00 to 99

Note: This is format and is not precision

Where Used: S2F18,S2F31

TOTSMP **Format:54**

Total sample count

Where Used: S2F23,S6F1

TRID **Format:54**

Trace request ID

Where Used: S2F23,S6F1

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UNITS **Format:20**

Units and Note

Where Used: S1F12,S2F30,S2F48

UPPERDB **Format:11,20,3(),4(),5()**

A variable limit attributes which defines the upper boundary of the dead band of a limit. The value applies to a single limit(LIMITID) for a specified VID. Thus, UPPERDB and LOWERDB as a pair define a limit.

Where Used: S2F45,F2F48

V **Format:0,10,11,20,21,3(),4(),5()**

Variable Data

Where Used: S6F11,S6F16,S6F20

VID **Format:52.54**

Variable ID

Where Used: S2F33,S2F45,S2F46,S2F47,S2F48

VLAACK **Format:10**

Variable Limit Attribute Acknowledge Code, 1byte
0 = Acknowledge, command will be performed
1 = Limit attribute definition error
2 = Cannot perform now
65 = System Error

Where Used: S2F46

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7-2. Item Format Code

コード Octal	Format	Description
00	L	LIST (length in elements)
10	B	Binary
11	BOOLEAN	Boolean
20	A	ASCII
21	JIS-8	JIS-8
30	I8	8bytes integer (signed)
31	I1	1byte integer (signed)
32	I2	2bytes integer (signed)
34	I4	3bytes integer (signed)
40	F8	8bytes floating point
44	F4	4bytes floating point
50	U8	8bytes integer (unsigned)
51	U1	1bytes integer (unsigned)
52	U2	2bytes integer (unsigned)
54	U4	4bytes integer (unsigned)

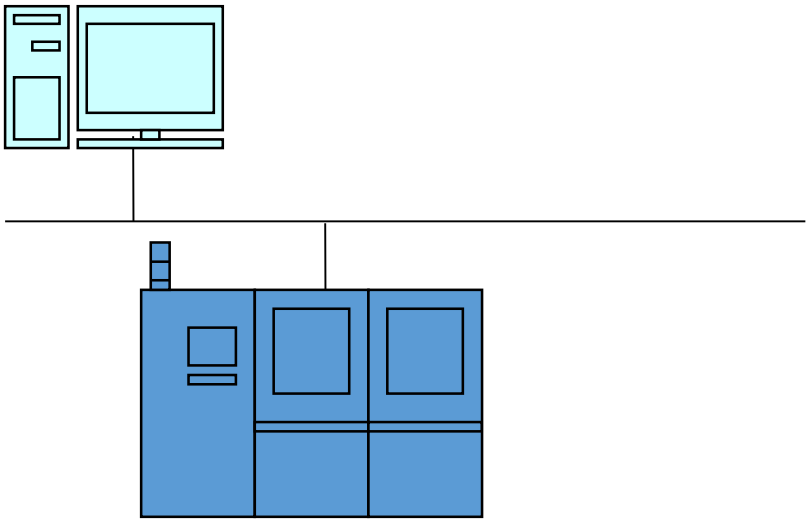
※This system do not have Format Code 22.

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8. Setup

8-1. LAN cable

Please connect the equipment PC and HOST computer.
At this time, use a LAN adapter different from the equipment network.
Please provide the LAN cable by customer.



8-2. HSMS Setting

Please setup the HSMS in the EQPanel(GUI).
Refer EQGUI in “GEM Document(Chapter3)” for the detail.

8-3. GEM Setting

Please setup the GEM from the HOST computer.
Refer the Appendix in “GEM Document(Chapter2)” for the SVIDs and CEIDs using by S2F33/F35/F37.

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